Quantum-rational sums and holonomic boundary certificates

Abstract

Let Δ be a rational element of the quantum plane $YX = tXY$, expanded as a t-adic series, and let $S(\Delta) \in \mathbb{Q}[[t]]$ be the sum of its coefficients at $X = Y = 1$. We certify $S(\Delta) = 0$ by a *boundary identity* $\Delta = (1-X)a + b(1-Y)$. The canonical boundary b always exists as a formal series and always satisfies a linear t-difference equation; the whole difficulty is to prove that it *converges*. We therefore split a certificate into two independent parts: an algebraic identity that pins down b, and a finite proof that its coefficients tend to 0. For the first part we give operator certificates rational in the shift T, found by solving a linear q-difference equation given by the least annihilator of the marginal; those without a pole at $T = 0$ prove convergence as well, and give $S(\Delta)$ exactly. For infinite product identities a pole at $T = 0$ is forced. For the second part we show that decay along a single arithmetic progression suffices, and give two finite tests: uniform contraction, and a channel test with *dilation witnesses*. Euler's odd–distinct partition identity is proved this way. Since only the sum matters, each block of the input may be replaced by other systems with the same sum: its second marginal, obtained by exchanging X and Y, and the second marginal of a canonical realization. For partition generating functions these count by largest part and by number of parts. Searching over these readings proves all 79 differences within four families of realizations with a common sum, and all 813 differences within 497 lists of automata with a common growth series: most by a rational certificate without pole, and the infinite product identities by the channel test. A sum side, whose partial sums are not q-hypergeometric, is handled by channels up to a remainder that tends to 0: the channels are hypergeometric elements of the module of the partial sums, obtained from the adjoint recurrence. With this, all 3237 differences within a second set of 49 lists are proved except the 29 that involve three automata whose series equal the others by Gauss's theta-function identity. A short Julia implementation built on the PBW algebras of Oscar accompanies the paper.

1 Overview

The problem. Write $K = \mathbb{Q}(t)$ and let $A = K\langle X, Y\rangle/(YX - tXY)$ be the quantum plane. A rational element Δ of A that is regular at $t = 0$ has a t-adic expansion $\Delta = \sum_k t^k H_k(X, Y)$ in normally ordered polynomials H_k (all X before all Y). Its *sum* is

$$S(\Delta) = \sum_k t^k H_k(1, 1) \in \mathbb{Q}[[t]].$$

Many q-series identities have the form $S(\Delta) = 0$ for a difference $\Delta = h - h'$ of two rational generating functions; Section 8 treats Euler's identity in this form.

Boundary certificates. If $\Delta = (1-X)a + b(1-Y)$ with a, b in the t-adic completion, then $S(\Delta) = 0$ by telescoping. Conversely, whenever $S(\Delta) = 0$ there is a unique such pair with b free of X (Proposition 3.2). The function $b = b(Y)$ is the generating function of the partial sums of the marginal $C = \rho(\Delta)$, so

$$b \text{ converges} \iff S(\Delta) = 0.$$

A *complete* certificate consists of

1. an **identity** that determines b by finite data, and

2. a finite **convergence proof** that the coefficients of b tend to 0.

Part 1 is always possible: b is holonomic (Theorem 4.2). Part 2 is the real content; no complete method for it is known, and we do not claim one.

Results.

- *Rational operator certificates* (Section 5), with coefficients rational in T, come from the rational solutions of a q-difference equation (Abramov's method), best formed from the least annihilator of the marginal (Proposition 5.5). Without a pole at $T = 0$ they deliver both parts at once and compute $S(\Delta)$; their existence is decidable.
- *Poles at* $T = 0$ (Section 6) are inverse shifts: such certificates deliver Part 1 only, and the order of the pole is the valuation that the convergence proof must make up. For identities between infinite products the pole is forced (Proposition 6.2).
- *One progression suffices* (Corollary 3.4): the partial sums are t-adically Cauchy at a rate read off from the input, so decay along one arithmetic progression proves $S(\Delta) = 0$, and then yields an explicit linear decay rate for all coefficients of b.
- *Convergence tests* (Section 7): uniform matrix contraction, and a channel test based on decimation and dilation witnesses, up to a finite boundary factor. The latter proves Euler's identity (Section 8).
- *Remainders* (Section 7.5): partial sums that are not a sum of q-hypergeometric terms may be one up to a regular combination of themselves that tends to 0. The terms are hypergeometric elements of their module, dual to the q-hypergeometric solutions of the adjoint recurrence, and the remainder is found by linear algebra on Laurent coefficients. This proves identities with a sum side.
- *Readings* (Section 9): a block of the marginal system may be replaced by its second marginal (exchange of X and Y) or by the second marginal of a canonical realization of its marginal, without changing S. Searching over readings proves all 79 differences within four families of realizations with a common sum (Section 9.5), all 813 differences within 497 lists of automata with a common growth series, and, with remainders, all differences within a second set of 49 lists except those that need Gauss's identity (Section 9.6).

Route	Part 1 (identity)	Part 2 (convergence)
Rational certificate, no pole at $T = 0$	decidable	automatic
Rational certificate, pole of order s	decidable	must gain s per index
Vector telescoper	bounded ansatz	contraction or channel test

2 Algebras and series

2.1 Completions

Let $\widehat{K} = \mathbb{Q}((t))$ with valuation $\nu = \nu_t$, $\nu(0) = +\infty$. Put

$$\begin{aligned} &A_0 = \mathbb{Q}[t]\langle X, Y\rangle/(YX - tXY), \quad A = K \otimes A_0, \quad \widehat{A} = \varprojlim_N A_0/t^N A_0, \\ &B_0 = \mathbb{Q}[t][Y], \qquad\qquad\quad B = K[Y], \qquad \widehat{B} = \varprojlim_N B_0/t^N B_0 = \mathbb{Q}[Y][[t]]. \end{aligned}$$

Elements of $\widehat{A}$ are series $\sum_{k\geq 0} t^k H_k(X, Y)$ with $H_k \in \mathbb{Q}[X, Y]$ written in normal order; multiplication is the quantum product. The algebras $B_0, B, \widehat{B}$ are the X-free parts. (Completion is taken before inverting t: since t is a unit in A, the t-adic completion of A itself is 0.)

Two larger series spaces carry convergence and its failure:

$$B^{\mathrm{res}} = \widehat{B}[t^{-1}] \qquad \text{and} \qquad B^{\mathrm{for}} = \widehat{K}[[Y]].$$

For $v = \sum_n v_n Y^n \in B^{\mathrm{for}}$,

$$v \in B^{\mathrm{res}} \iff \nu(v_n) \to +\infty, \qquad v \in \widehat{B} \iff \nu(v_n) \to +\infty \text{ and } \nu(v_n) \geq 0 \text{ for all } n. \tag{1}$$

Thus $B \subset \widehat{B} \subset B^{\mathrm{res}} \subset B^{\mathrm{for}}$ and $K[[Y]] \subset B^{\mathrm{for}}$, but $K[[Y]] \not\subset B^{\mathrm{res}}$. Evaluation at $Y = 1$, $v \mapsto \sum_n v_n$, is defined on B^{res} (a t-adically convergent sum) and maps $\widehat{B}$ to $\mathbb{Q}[[t]]$; it is not defined on B^{for}. *Formal existence in B^{for} and convergence in B^{res} are different assertions*, and the paper is organized around this distinction.

2.2 The operator algebra and holonomic series

Let Y act on B^{for} by multiplication and T by $Tv(Y) = v(tY)$. Since $TY = tYT$, this is an action of

$$\mathcal{O} = K\langle Y, T\rangle/(TY - tYT), \qquad \Big(\sum_j h_j(Y)T^j\Big) \cdot v = \sum_j h_j(Y)\, v(t^j Y).$$

We write elements of $\mathcal{O}$ in normal order $Y^b T^a$.

Definition 2.1. A series $v \in B^{\mathrm{for}}$ is *holonomic* if $H \cdot v = 0$ for some $0 \neq H \in \mathcal{O}$; equivalently, the $K(Y)$-span of $\{T^j v\}_{j \geq 0}$ in $\widehat{K}((Y))$ is finite-dimensional. Write B^{hol} for the set of holonomic series.

The coefficient field is $K(Y)$, not $\widehat{K}(Y)$; this is q-holonomicity in Y with $q = t$ (cf. [1]), not differential finiteness in t.

Proposition 2.2. *B^{hol} is a $\widehat{K}$-subalgebra of B^{for} containing $\widehat{K}[Y]$. It is stable under $\mathcal{O}$, under $T^{-1}v = v(Y/t)$, and under multiplication by elements of $K(Y)$ whenever the product lies in B^{for}.*

Proof. Let V_v be the shift span of v. The shifts of $v + w$ lie in $V_v + V_w$, and those of vw in the span of products of bases of V_v and V_w. On the finite-dimensional space V_v the map T is injective and semilinear for the automorphism $a(Y) \mapsto a(tY)$ of $K(Y)$, hence bijective; so $T^{-1}v \in V_v$ and $V_{T^{-1}v} = V_v$. The other statements are immediate. □

B^{hol} is *not* a field: Appendix A gives $g \in B^{\mathrm{hol}} \cap \widehat{B}$ with $g^{-1} \in \widehat{B} \setminus B^{\mathrm{hol}}$. Its fraction field $F^{\mathrm{hol}} = \mathrm{Frac}(B^{\mathrm{hol}}) \subset \widehat{K}((Y))$ consists of ordinary quotients of commuting series; field membership carries no difference equation. Note also that B^{hol} imposes no convergence: the relevant classes are $B^{\mathrm{hol}} \cap B^{\mathrm{res}}$ and $B^{\mathrm{hol}} \cap \widehat{B}$.

2.3 Operator fractions are not series

The skew field $D_{\mathcal{O}} = \mathrm{Frac}(\mathcal{O})$ consists of noncommuting operator fractions. It does not act on B^{for}: the nonzero operator $T - 1$ kills the series 1, whereas every nonzero element of $D_{\mathcal{O}}$ is invertible. For instance $(1-tT)\cdot v = 1$ has the unique solution $v = (1-t)^{-1}$, while $(1-tT)^{-1} \in D_{\mathcal{O}}$ is not a scalar; its geometric expansion acts on $\widehat{B}$ and gives $(1 - tT)^{-1} \cdot 1 = (1 - t)^{-1}$. We use $D_{\mathcal{O}}$ only for symbolic elimination, and every inverse applied to a series is given by an explicitly convergent expansion such as (6) below. We write $\varepsilon(H) = H \cdot 1 \in K[Y]$; it is linear but not multiplicative ($\varepsilon(TY) = tY \neq Y = \varepsilon(T)\varepsilon(Y)$).

3 From a realization to the marginal system

3.1 Realizations and the marginal

A *right realization* of Δ is

$$wQ = r, \qquad \Delta = we, \qquad Q \in \mathrm{Mat}_n(A),\ r \in A^{1\times n},\ e \in A^{n\times 1}, \tag{2}$$

with Q invertible over the fraction skew field of A. We always assume the *normalization*

$$Q \equiv I \pmod{t}, \qquad r, e \text{ regular at } t = 0 \tag{3}$$

coefficientwise in the normal basis, and select $w = rQ^{-1} \in \widehat{A}^{1\times n}$ by geometric inversion. (Convergence of Δ alone does not imply convergence of all coordinates w_i of an arbitrary realization.)

The *marginal* and the *sum* are the linear maps

$$\rho\Big(\sum_k t^k H_k(X,Y)\Big) = \sum_k t^k H_k(1,Y) \in \widehat{B}, \qquad S(v) = \rho(v)(1) \in \mathbb{Q}[[t]].$$

ρ is not multiplicative: $\rho(YX) = tY \neq \rho(Y)\rho(X)$.

Lemma 3.1 (intertwining)**.** *Let $\Phi : A \to \mathcal{O}$ be the anti-isomorphism $\Phi(X^aY^b) = Y^bT^a$. For $w \in \widehat{A}[t^{-1}]$ and $q \in A$,*

$$\rho(wq) = \Phi(q) \cdot \rho(w).$$

Moreover $\ker \rho|_{\widehat{A}} = (1-X)\widehat{A}$.

Proof. For $w = X^iY^j$, $q = X^aY^b$: $wq = t^{ja}X^{i+a}Y^{j+b}$, so $\rho(wq) = t^{ja}Y^{j+b} = Y^bT^a \cdot Y^j$. The kernel statement is polynomial division by $1 - X$ in each t-degree. □

In other words, $A/(1-X)A \cong B$ as a right A-module, and Φ turns the right action into the left $\mathcal{O}$-action. Applying the lemma to (2) gives the *compiled triple*

$$L_{ji} = \Phi(Q_{ij}), \qquad f_j = \rho(r_j) \in B, \qquad d_i = \Phi(e_i), \tag{4}$$

and the *marginal system*

$$L \cdot m = f, \qquad C := \rho(\Delta) = d \cdot m, \qquad m_i = \rho(w_i) \in \widehat{B}. \tag{5}$$

By (3), $L = I - N$ where every coefficient of N has positive valuation, and the selected solution is

$$m = \sum_{j\geq 0} N^j \cdot f, \tag{6}$$

the unique solution in $\widehat{B}^n$. All later objects are computed from the triple (L, f, d) alone.

3.2 The canonical boundary

Proposition 3.2. *For $\Delta \in \widehat{A}$ there exist $a, b \in \widehat{A}$ with*

$$\Delta = (1-X)a + b(1-Y) \tag{7}$$

if and only if $S(\Delta) = 0$. In that case there is exactly one such pair with $b \in \widehat{B}$, namely

$$b = \frac{C}{1-Y}, \qquad a = (1-X)^{-1}\big(\Delta - C(Y)\big).$$

Proof. Telescoping gives necessity. Write $C = \sum_k t^k C_k(Y)$. The quotient $C/(1-Y)$ has polynomial t-coefficients iff $C_k(1) = 0$ for all k, i.e. iff $S(\Delta) = 0$; then $\Delta - C$ has zero marginal and is divisible by $1 - X$ (Lemma 3.1). For an arbitrary pair, applying ρ replaces b by $\rho(b) \in \widehat{B}$ and division reconstructs a; uniqueness follows from injectivity of multiplication by $1 - X$ and by $1 - Y$. □

Write $C = \sum_n c_n Y^n$ and $\beta_n = \sum_{j \le n} c_j$. Since $1 - Y$ is a unit in B^{for}, the *formal boundary* always exists:

$$b = \frac{C}{1-Y} = \sum_n \beta_n Y^n \in B^{\mathrm{for}}, \qquad b \in \widehat{B} \iff \beta_n \to 0 \iff S(\Delta) = 0. \tag{8}$$

Under (3) all c_n, β_n lie in K and are integral.

3.3 An unconditional rate

Choose integers

$$F_0 \ge \deg_Y f, \qquad E \ge \deg_Y d, \qquad D \ge \max\{1, \deg_Y L\}, \qquad \alpha = F_0 + E.$$

Each factor N in (6) raises t-order by at least one and Y-degree by at most D, so

$$\deg_Y [t^k]\, m_i \le F_0 + Dk, \qquad \deg_Y [t^k]\, C \le \alpha + Dk. \tag{9}$$

Hence $\nu(c_n) \ge \lceil (n - \alpha)/D \rceil$, and with

$$k_\Delta(n) = \max\Big\{0, \Big\lceil \frac{n + 1 - \alpha}{D} \Big\rceil\Big\} \tag{10}$$

we obtain:

Proposition 3.3. *For every normalized input and all $n' \ge n \ge 0$,*

$$\nu(S(\Delta) - \beta_n) \ge k_\Delta(n), \qquad \nu(\beta_{n'} - \beta_n) \ge k_\Delta(n).$$

Consequently:

1. *if $\nu(\beta_n) < k_\Delta(n)$ for some n, then $S(\Delta) \ne 0$, and the leading term of β_n is the leading term of $S(\Delta)$;*
2. *$S(\Delta) = 0$ if and only if $\nu(\beta_n) \ge k_\Delta(n)$ for all n.*

Corollary 3.4 (one progression suffices)**.** *If $\nu(\beta_{n_j}) \to \infty$ along some subsequence $n_j \to \infty$, then $S(\Delta) = 0$, $b \in \widehat{B}$, and $\nu(\beta_n) \ge k_\Delta(n)$ for every n.*

Proof. $S(\Delta) = \lim_n \beta_n = \lim_j \beta_{n_j} = 0$; then apply Proposition 3.3(2). □

So a convergence proof need only show decay along one arithmetic progression, at any rate; the uniform linear rate k_Δ then comes for free. Note that k_Δ does not by itself decide anything: checking item 2 for all n *is* the zero problem.

4 The boundary is always holonomic

Lemma 4.1 (elimination)**.** *Let $G \in \mathrm{Mat}_s(\mathcal{O})$ be nonsingular over $D_{\mathcal{O}}$ with T-degree at most D, and fix a column index k. There are $U \in \mathcal{O}^{1\times s}$ and $0 \neq H \in \mathcal{O}$ with $UG = H\, e_k^{\mathsf{T}}$ and $\deg_T U \leq (s-1)D$.*

Proof. Seek $U_j = \sum_{i\leq v} u_{ji}(Y)T^i$ over $K(Y)$ annihilating all columns except the k-th. There are $s(v+1)$ unknowns and at most $(s-1)(v+D+1)$ linear equations over $K(Y)$; for $v = (s-1)D$ a nonzero solution exists. The surviving entry H is nonzero because G is nonsingular. Clear a common left denominator. □

Theorem 4.2. *For every normalized input, $m_i, C \in B^{\mathrm{hol}} \cap \widehat{B}$ and $b \in B^{\mathrm{hol}}$, with effectively computable annihilators. Moreover $S(\Delta) = 0 \iff b \in B^{\mathrm{hol}} \cap \widehat{B}$. Thus holonomicity imposes no further existence condition; it provides no convergence proof.*

Proof. Adjoin the constant $u = 1$, selected by $(T-1)u = 0$, $u(0) = 1$. Then

$$G \cdot \begin{pmatrix} m \\ u \\ b \end{pmatrix} = 0, \qquad G = \begin{pmatrix} L & -f & 0 \\ 0 & T-1 & 0 \\ -d & 0 & 1-Y \end{pmatrix}$$

with G nonsingular. Lemma 4.1 gives $UG = (0, \ldots, 0, H)$, a finite identity in $\mathcal{O}$; applying it to the selected solution gives $H \cdot b = 0$. Other columns give annihilators of the m_i, and Proposition 2.2 gives $C = d \cdot m \in B^{\mathrm{hol}}$. The last statement is (8). □

No inverse from $D_{\mathcal{O}}$ is applied to a series here, so there is no conflict with Section 2.3. An annihilator alone does not select b: the selected m, $u = 1$ and $(1-Y)b = dm$ do.

Example 4.3. $\Delta = 1$ gives $b = (1-Y)^{-1} \in B^{\mathrm{hol}} \setminus B^{\mathrm{res}}$, described finitely by $\big((1-tY)T - (1-Y)\big)b = 0$, $b(0) = 1$. Every coefficient has valuation 0: a finite description is not a convergence proof.

The component a has a similar description by replacing Φ with $\Psi(X^aY^b) = x^aY^bT^a$ over $\mathcal{O}[x]$; no separate search for a is needed.

4.1 Identically vanishing marginals

An important special case is $C \equiv 0$. It occurs whenever $\Delta = h - h'$ for two realizations of the same series, or more generally of series with the same marginal. Then $b = 0$ and there is nothing to converge, yet a boundary certificate may still need a shift (Example 4.6). We certify $C = 0$ directly by an annihilator. Write $H = \sum_b Y^b H_b(T)$ for $H \in \mathcal{O}$.

Proposition 4.4 (zero certificate)**.** *Let $0 \neq H \in \mathcal{O}$ and $R \in \mathcal{O}^{1\times n}$ satisfy*

$$H\, d = R\, L, \qquad \varepsilon(Rf) = 0. \tag{11}$$

Let b_0 be least with $H_{b_0} \neq 0$, and $E = \{k \geq 0 : H_{b_0}(t^k) = 0\}$. Then E is finite and computable, and $C = 0$ if and only if $c_k = 0$ for all $k \in E$.

Proof. $H \cdot C = (Hd) \cdot m = R \cdot (L \cdot m) = R \cdot f = \varepsilon(Rf) = 0$. Taking the coefficient of Y^{k+b_0},

$$H_{b_0}(t^k)\, c_k = -\sum_{b > b_0} H_b(t^{k+b_0-b})\, c_{k+b_0-b}.$$

By the unique-minimal-term argument of Section 7, the polynomial $H_{b_0}(z)$ is nonzero at $z = t^k$ for all k beyond an explicit bound, so E is finite. If $c_k = 0$ for $k \in E$, induction on k gives $c = 0$. □

Proposition 4.5. *For every normalized input, a pair (H, R) satisfying (11) exists. Consequently it is decidable whether $C \equiv 0$; in particular, equality of the marginals of two realizations is decidable.*

Proof. The matrix $G' = \left(\begin{smallmatrix} L & -f \\ 0 & T-1 \end{smallmatrix}\right)$ from the proof of Theorem 4.2 is invertible over $D_{\mathcal{O}}$, so $(d, 0) = VG'$ for some row V over $D_{\mathcal{O}}$. A common left denominator $H \neq 0$ of its entries (Ore condition) gives $U = HV = (R, R_u)$ over $\mathcal{O}$ with $UG' = H(d, 0)$, that is, $RL = Hd$ and $Rf = R_u(T-1)$; applying the last identity to 1 gives $\varepsilon(Rf) = 0$. For the decision procedure, search (H, R) by increasing degree, which terminates because a solution exists. Then compute E, and compute c_k for $k \in E$. □

The equations (11) are linear and homogeneous in the coefficients of H and R. With a degree ansatz they are a kernel computation over K.

Example 4.6 (two realizations)**.** For the distinct-parts series $h' - 1$ of Section 8, the PBW realization gives the marginal system $L = 1 - tY - t^2YT$, $f = 1$, $d = tY$. A two-state automaton gives $L' = \left(\begin{smallmatrix} 1-tY & -tY \\ -tT & 1 \end{smallmatrix}\right)$, $f' = (1, 0)$, $d' = (tY, 0)$. Their difference is the block system $\operatorname{diag}(L, L')$, $(1, 1, 0)$, $(tY, -tY, 0)$ with $C \equiv 0$. It has no polynomial certificate. A certificate with a pole of order 1 at $T = 0$ exists, but its output is a cancelling combination of divergent series beyond the channel test. Proposition 4.4 applies at once:

$$H = 1 - tY - tYT, \qquad R = (tY,\ -tY,\ -t^2Y^2), \qquad E = \emptyset.$$

For example $H \cdot tY = tY - t^2Y^2 - t^3Y^2T = tY \cdot L$.

5 Rational operator certificates

Rational coefficients act on series. Since $T \cdot Y^n = t^n Y^n$, a rational function $r \in K(T)$ acts on $v = \sum v_n Y^n$ coefficientwise,

$$r(T) \cdot v = \sum_n r(t^n)\, v_n\, Y^n,$$

at every n where r has no pole at t^n. By the unique-minimum argument of Section 7, a nonzero polynomial vanishes at $z = t^n$ for finitely many computable n only. So operators $\sum_b Y^b r_b(T)$ with coefficients in $K(T)$ act on all coefficients of large index. This is the sense in which certificates may be rational. (Fractions of operators in general do not act, Section 2.3; fractions in T alone do.)

Definition 5.1. A *rational certificate* is a pair of rows P, R with entries in $K(T)[Y]$ with

$$d = (1 - Y)P + RL. \tag{12}$$

It is *polynomial* if R and P have entries in $\mathcal{O}$.

Proposition 5.2. *Let (P, R) be a rational certificate and $q_n = [Y^n](P \cdot m)$. Then $c_n = q_n - q_{n-1} + [Y^n](R \cdot f)$ for all large n, so $\beta_n - q_n = \kappa$ is eventually constant. If no entry of P has a pole at $T = 0$, then $q_n \to 0$ and*

$$S(\Delta) = \kappa.$$

If moreover no coefficient has a pole at any t^n, then $\kappa = (R \cdot f)(1)$ and $b = P \cdot m + (R \cdot f)/(1 - Y)$.

Proof. Applying (12) to m gives $C = (1 - Y)P \cdot m + R \cdot f$ at every index where the coefficients involved are defined, and $R \cdot f$ is a polynomial. If $p \in K(T)$ has no pole at 0, then $\nu p(t^j)$ is eventually constant, so $\nu(q_n) \geq \min_b \nu(m_{n-b}) - \text{const} \to \infty$. □

So a rational certificate without pole at $T = 0$ is a complete proof of $S(\Delta) = 0$, and otherwise computes $S(\Delta) \in K$ exactly. Denominators that do not vanish at $T = 0$ are harmless: $1/q(t^n)$ has eventually constant valuation.

Reduction. Let Λ be evaluation at $Y = 1$ in normal order, $\Lambda(\sum Y^b r_b(T)) = \sum r_b(T)$, a map $K(T)[Y] \to K(T)$ with kernel $(1-Y)K(T)[Y]$. Every row decomposes as $R = R_0 + (1-Y)V$ with R_0 free of Y, and replacing (R, P) by $(R_0, P + VL)$ preserves (12). Hence a rational certificate exists iff

$$\Lambda(RL) = \Lambda(d), \qquad R \in K(T)^{1\times n}, \tag{13}$$

and P is recovered by exact left division by $1 - Y$: if $\Lambda(E) = 0$ for $E = \sum Y^b e_b(T)$, then $E = (1-Y)P$ with $P = \sum Y^b p_b$, $p_b = \sum_{b' \le b} e_{b'}$. In coordinates $z \leftrightarrow T$, with $L_{ji} = \sum \ell_{ji,ba} Y^b T^a$ and $R_j = R_j(z)$, (13) reads

$$\sum_{j,a,b} \ell_{ji,ba}\, z^a R_j(t^b z) = \Lambda(d_i)(z) \qquad (1 \le i \le n), \tag{14}$$

because $\Lambda(r(T)Y^b T^a) = r(t^b z)z^a$: a nonsingular linear t-difference system for R (the transpose of L under $Y \mapsto S$, $T \mapsto z$, $Sz = tzS$).

Rational solutions. Consider a scalar equation $\sum_{b=0}^{B} a_b(z)\, y(t^b z) = r(z)$ with polynomial coefficients, $a_0, a_B \neq 0$ (Lemma 4.1 applied to (14) gives one for each R_j). Its rational solutions are found as follows [4].

1. *Denominator.* Let u be an irreducible factor of the denominator, $u \neq z$, and consider the chain of factors $u(t^i z)$ of the denominator. Its first element divides $a_0(z)$ and its last divides $A(z) = a_B(t^{-B}z)$, since otherwise that factor could not cancel in the equation. So the denominator divides $z^k U(z)$, where U collects, for s from the largest dispersion down to 0, $d(z)d(t^{-1}z)\cdots d(t^{-s}z)$ with $d = \gcd(A(z), a_0(t^s z))$, removing d from A and $d(t^{-s}z)$ from a_0 at each step.
2. *Pole at* 0. If $y \sim cz^{-k}$, the lowest term of the left side is $cz^{v-k}\chi_0(t^{-k})$, where $v = \min_b \operatorname{ord}_z a_b$ and $\chi_0(W) = \sum_b [z^v] a_b W^b$. So $k = v - \operatorname{ord}_z r$ or $\chi_0(t^{-k}) = 0$: finitely many k.
3. *Numerator.* $N = z^k U y$ is a polynomial solution of an equation of the same kind, of degree bounded by Lemma 5.3. Its coefficients solve a linear system over K, and the condition "no pole at 0" is linear in them.

Lemma 5.3 (degree bound)**.** *Let $\sum_{j=0}^{v} a_j(z) R(t^j z) = g(z)$ with $a_j, g \in K[z]$, not all $a_j = 0$. Put $D = \max_j \deg a_j$, $\chi(W) = \sum_j [z^D] a_j W^j$, and let E bound the t-degrees of the coefficients of χ after clearing denominators. Every polynomial solution satisfies*

$$\deg R \le \max\{0, E, \deg g - D\} \qquad (\textit{omit } \deg g \textit{ if } g = 0).$$

Proof. If $\deg R = N$, the coefficient of z^{N+D} on the left is $\chi(t^N)$ times the leading coefficient of R. For $N > E$ the terms of $\chi(t^N)$ occupy disjoint t-degree ranges, so $\chi(t^N) \neq 0$; this forces $N + D \le \deg g$. □

Theorem 5.4. *Existence of a rational certificate, and of one without pole at $T = 0$, is decidable; such a certificate yields $S(\Delta)$ exactly. No bound in Y is needed.*

Polynomial solution bounds for q-difference systems are also treated in [2]. The implementation uses the scalar form of (13) given by the annihilator: if $H \cdot C = g$, the 1×1 identity $1 = (1 - Y)P + R(T)H$ gives $C = (1 - Y)P \cdot C + R \cdot g$, and the condition is

$$\Lambda(R\,H) = \sum_b h_b(t^b z)\, R(t^b z) = 1, \qquad H = \sum_b h_b(T) Y^b \tag{15}$$

(coefficients on the left), a scalar equation of order $\deg_Y H$. Here $b = P \cdot C$ is an image of C itself; the system form can also succeed when b is an image of the auxiliary series but not of C, and its polynomial case is searched as well.

The least annihilator. The scalar form depends on H, and the annihilator obtained from the system need not be the best one. The operators $A \in K(T)[Y]$ for which the coefficients of $A \cdot C$ vanish at all large indices form a left ideal of the Ore algebra $K(T)[Y]$, which is principal; let A_C be a generator. A certificate $b = P \cdot C + (\text{polynomial})/(1-Y)$ exists iff $1-(1-Y)P$ lies in this ideal, that is, by the reduction above, iff (15) has a rational solution for $H = A_C$. A solution for a multiple $H = QA_C$ gives one for A_C (reduce RQ modulo $(1-Y)$), but not conversely. The annihilator from the system is least for the class of d, not for C, and A_C is often smaller: when f excites only part of the system, and typically for a difference $C = C_1 - C_2$ of two series that share a component. If C is a polynomial, $A_C = 1$. A candidate A is easy to guess from $c_0, \dots, c_N$ by linear algebra; it is then proved as follows.

Proposition 5.5 (a smaller annihilator)**.** *Let $H \cdot C = g$ with g a polynomial, and let $H = QA$ with $A \in \mathcal{O}$ and $Q \in K(T)[Y]$. Choose $D \in K[T]$ with $\widetilde{Q} = DQ \in \mathcal{O}$, let g_A be the part of $A \cdot C$ of degree less than n_0, and let E be the exceptional set of $\widetilde{Q}$ (Proposition 4.4). If $\widetilde{Q} \cdot g_A = D \cdot g$ and the coefficients of $A \cdot C$ vanish at the indices $n_0 \le n \le \max E$, then $A \cdot C = g_A$.*

Proof. The series $u = A \cdot C - g_A$ satisfies $\widetilde{Q} \cdot u = D \cdot (H \cdot C) - \widetilde{Q} \cdot g_A = D \cdot g - D \cdot g = 0$. Its coefficients vanish below n_0 by the choice of g_A, and up to $\max E$ by assumption. The induction in the proof of Proposition 4.4 gives $u = 0$. □

Example 5.6 (a rational denominator)**.** $\Delta = (1-t)(1-tX)^{-1}(X-1)$ compiles to $L = 1 - tT$, $f = 1 - t$, $d = T - 1$. Here (13) reads $R(z)(1-tz) = z - 1$, whose solution $R = (z-1)/(1-tz)$ is not polynomial (nor Laurent) but has no pole at $z = 0$ or at any $z = t^n$: a complete certificate, with $S(\Delta) = R(1)(1-t) = 0$. (In fact $C = 0$ and $\Delta = (1-X)a$; Proposition 4.4 also applies, with $H = 1 - tT$, $R = T - 1$ and $E = \emptyset$.) For a nonzero marginal, the system $L = \left(\begin{smallmatrix} 1-tY & 0 \\ -t & 1-tT^2 \end{smallmatrix}\right)$, $f = (1, 0)$, $d = (0, 1-Y)$ has $\beta_n = t^{n+1}/(1-t^{2n+1})$, and the scalar certificate (15) gives $b = P \cdot C$ with $P = (t - T^2)/\big((t-1)(1+T^2)\big)$.

6 Poles at $T = 0$ and shifted unknowns

A denominator prime to T is harmless (Proposition 5.2). The factor T is not: $T^{-1}v = v(Y/t)$ preserves B^{for} and B^{hol} but not B^{res},

$$(1-tY)^{-1} \in \widehat{B}, \qquad T^{-1}(1-tY)^{-1} = (1-Y)^{-1} \notin B^{\text{res}}.$$

Fix $s \ge 0$ and change unknowns:

$$\mu = T^{-s}m, \quad \widetilde{L} = T^{-s}LT^s, \quad \widetilde{f} = T^{-s} \cdot f, \quad \widetilde{d} = dT^s, \qquad \text{so} \qquad \widetilde{L}\mu = \widetilde{f}, \quad C = \widetilde{d}\mu. \tag{16}$$

Since $T^{-s}YT^s = t^{-s}Y$, conjugation only rescales coefficients: $\widetilde{L}$ is again polynomial. The μ_i lie in B^{hol} but need not converge.

Proposition 6.1. *Rational certificates whose entries have poles of order at most s at $T = 0$ correspond bijectively, via $P_s = PT^s$, $R_s = RT^s$, to rational certificates $\widetilde{d} = (1-Y)P_s + R_s\widetilde{L}$ of the shifted system without pole at $T = 0$. Either form proves*

$$b = P_s \cdot \mu + (R_s \cdot \widetilde{f})/(1-Y) \in B^{\text{hol}}, \qquad (1-Y)b = C,$$

i.e. it identifies the canonical formal boundary. It does not *prove convergence.*

Proof. Multiply (12) on the right by T^s; left division by $1 - Y$ does not change the order at $T = 0$. □

What the shift costs. If $g = T^s b = \sum \gamma_n Y^n$ then $\gamma_n = t^{sn}\beta_n$. A convergent expression for g only gives $\nu(\gamma_n) \to \infty$; one needs $\nu(\gamma_n) \geq sn + k(n)$ with $k(n) \to \infty$. Inverse shifts move difficulty from the identity to the estimate; they do not remove it. For identities between infinite products the pole at $T = 0$ cannot be avoided:

Proposition 6.2 (forced pole)**.** *Suppose that along some progression $n = pm + r$ the auxiliary sequences (or C, in the scalar form) are sums of q-hypergeometric terms whose valuations tend to infinity, while the decomposition of β into terms with pairwise non-similar ratios (Section 7.4) contains a term of bounded valuation whose ratio is not similar to* 1*. Then every rational certificate has a pole at $T = 0$.*

Proof. Without a pole, $\beta_{pm+r} - \kappa$ is a combination of finitely many shifts of the auxiliary sequences, with coefficients $r(t^m)$ of bounded valuation for large m. A q-hypergeometric term times such a coefficient is a term with a similar ratio, of valuation still tending to infinity. Grouping by similarity classes and using the uniqueness of the decomposition, every term of $\beta - \kappa$ would have valuation tending to infinity; the constant κ only affects the class of ratio 1. □

For Euler's identity, $\beta_{2m} = (t;t^2)_m^{-1} - (-t;t)_{2m}$ is a difference of two terms of valuation 0, while the auxiliary sequences have terms of valuation $2m$ (Section 8). So every certificate has a pole at $T = 0$; the limit $\beta_n \to 0$ is the infinite product identity, and its finite certificate is the dilation witness of Section 7, a rational identity $\theta(\zeta)\chi(\zeta) = \prod_i \chi(t^{\mu+i}\zeta^\lambda)$ between the ratios of the two products. The same holds for the 41 channel pairs of Section 9.5: in the readings used, the least pole at $T = 0$ of a rational certificate (15) has order 2.

7 Convergence certificates

7.1 Vector form of the output

For a normalized system with $L = \sum_{j=0}^{q} Y^j L_j(T)$, coefficient extraction gives

$$L_0(t^n)m_n = f_n - \sum_{j=1}^{q} L_j(t^{n-j})m_{n-j},$$

with $L_0(t^n)$ invertible over K. (The same holds for $\widetilde{L}$ and μ.) Stacking $m_n, \ldots, m_{n-q_i+1}$ (coordinate i needs only lag $q_i = \max_j \deg_Y L_{ji}$) gives, beyond a known index n_0,

$$v_{n+1} = A(t^n)v_n, \qquad \beta_n = \pi(t^n)\, v_n, \qquad A, \pi \text{ over } K(z). \tag{17}$$

An operator certificate yields π by coefficient extraction: $[Y^n]\, cY^jT^a\mu_i = c\, t^{a(n-j)}\mu_{i,n-j}$.

Alternatively, π can be found directly as a *vector telescoper*: a rational row with

$$\pi(tz)A(z) - \pi(z) = \lambda(z), \qquad \pi(t^N)v_N = \beta_N, \tag{18}$$

where $c_{n+1} = \lambda(t^n)v_n$; induction gives $\beta_n = \pi(t^n)v_n$ for $n \geq N$. A bounded ansatz for π is a linear system over K. Neither discovery route is a prerequisite for the other, and both feed the same convergence tests.

Valuations at $z = t^n$. For $0 \neq P(z) = \sum_i p_i z^i \in K[z]$ with least exponent i_0, the term i_0 is the unique minimum of $\nu(p_i) + in$ as soon as $n > \max_{i>i_0}(\nu p_{i_0} - \nu p_i)/(i - i_0)$; from then on $\nu P(t^n) = \nu(p_{i_0}) + i_0 n$. Applied to numerators and denominators, every valuation inequality below is a finite check.

7.2 Uniform contraction

Choose integers e_i, $D(z) = \mathrm{diag}(z^{e_i})$, $A_D(z) = D(tz)A(z)D(z)^{-1}$ and $\pi_D = \pi D^{-1}$. Suppose that for some $\ell \geq 1$ and N every entry of $A_D(t^{\ell-1}z)\cdots A_D(z)$ at $z = t^n$ has valuation ≥ 1 for $n \geq N$, and every entry of $\pi_D(t^n)$ has valuation $\geq h$. Then for $n \geq N$

$$\nu(\beta_n) \geq h + \min_{N \leq j < N+\ell} \nu\big(D(t^j)v_j\big) + \Big\lfloor \frac{n-N}{\ell} \Big\rfloor. \tag{19}$$

This forces decay for *every* initial vector, so it fails whenever decay depends on cancellation specific to the selected initial values (Remark 8.1).

7.3 Channels and dilation witnesses

Fix a period $p \geq 1$ and a residue r, and suppose the p-step product

$$A^{[p]}(x) = A(t^{p-1}x)\cdots A(tx)A(x)$$

is *diagonal*. Then along $n = pm + r$ each coordinate evolves by itself. Put $\zeta = t^m$, choose m_0 with $pm_0 + r \geq n_0$, and define the *channels*

$$G_k(m) = \pi_k(t^{pm+r})\, v_{pm+r,k}, \qquad g_k(\zeta) = \frac{\pi_k(t^{r+p}\zeta^p)}{\pi_k(t^r\zeta^p)}\, A^{[p]}_{kk}(t^r\zeta^p).$$

Then $G_k(m+1) = g_k(t^m)G_k(m)$ and $\beta_{pm+r} = \sum_k G_k(m)$ for $m \geq m_0$.

Proposition 7.1 (channel test)**.** *Let $\mathcal{K} = \{k : G_k(m_0) \neq 0\}$.*

(a) Decay. *If for each $k \in \mathcal{K}$ we have $\nu g_k(t^j) \geq 0$ for $j \geq m_0$ and $\nu g_k(t^j) \geq 1$ for $j \geq M$, then $\nu(\beta_{pm+r}) \geq m + \min_k \nu G_k(m_0) - M$.*

(b) Dilation. *Let $\mathcal{K} = \{k_1, k_2\}$ with $\nu g_{k_1}(t^j) \geq 0$ for $j \geq m_0$, and put $\theta = g_{k_2}/g_{k_1}$. Suppose*

(i) the virtual initial ratio is -1: $G_{k_2}(m_0) = -\prod_{j<m_0} \theta(t^j)\, G_{k_1}(m_0)$;

(ii) there are $\lambda \geq 2$ and $\chi \in K[\zeta]$ with $\chi(0) = 1$, integral coefficients, $\chi(1) \neq 0$, and

$$\prod_{i=0}^{\lambda-1} \chi(t^i\zeta^\lambda) = \theta(\zeta)\,\chi(\zeta). \tag{20}$$

Write $\chi - 1 = \sum_{i \geq i_1} \chi_i \zeta^i$ and $c = \min_i \nu\chi_i$. Then

$$\beta_{pm+r} = G_{k_1}(m)\Big(1 - \prod_{j=m}^{\lambda m-1} \chi(t^j)\Big), \qquad \nu(\beta_{pm+r}) \geq \nu G_{k_1}(m_0) + c + i_1 m.$$

In either case Corollary 3.4 gives $S(\Delta) = 0$.

Proof. (a) $\nu G_k(m) \geq \nu G_k(m_0) + \max(0, m - M)$. (b) First, $\prod_{j<m} \theta(t^j) = \prod_{j=m}^{\lambda m-1} \chi(t^j)$ for all $m \geq 0$, by induction: passing from m to $m+1$ multiplies the right side by $\prod_i \chi(t^{\lambda m+i})/\chi(t^m) = \theta(t^m)$ by (20) at $\zeta = t^m$; here $\chi(t^m) \neq 0$ since $\chi(1) \neq 0$ and $\chi(t^m) \equiv 1$ for $m \geq 1$. By (i) and the channel recursions, $G_{k_2}(m)/G_{k_1}(m) = -\prod_{j<m} \theta(t^j)$ for $m \geq m_0$, which gives the formula for β. Each $\chi(t^j) - 1$ has valuation $\geq i_1 j + c$, so $\nu\big(1 - \prod_{j=m}^{\lambda m-1} \chi(t^j)\big) \geq i_1 m + c$, and $\nu G_{k_1}(m) \geq \nu G_{k_1}(m_0)$. □

Finding the witness. Write $\theta = u/v$ in lowest terms. Comparing degrees in $v(\zeta)\prod_i \chi(t^i\zeta^\lambda) = u(\zeta)\chi(\zeta)$ shows

$$(\lambda^2 - 1)\deg\chi = \deg u - \deg v,$$

and comparing constant terms requires $\theta(0) = 1$. The coefficient of ζ^j on the left involves only coefficients of χ of degree $\le j/\lambda$, so χ is determined degree by degree from $\chi(0) = 1$. One computes it up to the predicted degree and checks (20) exactly. The identity says $\prod_{j\ge 0}\theta(t^j) = 1$, i.e. it is a finite certificate for an infinite product identity obtained by λ-section of the products $E_m = \prod_{j<m}\chi(t^j)$, namely $\prod_{j<m}\theta(t^j) = E_{\lambda m}/E_m$. The same recursion finds witnesses with an *offset* $\mu \in \mathbb{Z}$, i.e. solutions of

$$\prod_{i=0}^{\lambda-1} \chi(t^{\mu+i}\zeta^\lambda) = \theta(\zeta)\,\chi(\zeta); \tag{21}$$

if χ is a witness for θ, then $\chi(t^s\zeta)$ is a witness with offset $\mu = s(\lambda-1)$ for $\theta(t^s\zeta)$. So offsets arise when the channels are shifted in the index (Section 9.4).

The channel test is a restricted certificate class: it needs a diagonal decimation, at most two live channels, and a dilation witness. It is not a hypothesis on the input and carries no completeness claim.

7.4 Channels of the output sequence

The test above reads channels off a state space, so it depends on the realization. An automaton realizing the same series typically has redundant states: for the distinct-parts automaton of Example 4.6, the second coordinate is t^{n+1} times the first. Then no product $A^{[p]}$ is diagonal, and the test fails although the series is the same. We now describe a version of the test that depends only on the sequence (β_n).

A recurrence for β. As in the proof of Proposition 4.5 (without the condition on f), there are $H \neq 0$ and R with $Hd = RL$. Then $H \cdot C = \varepsilon(Rf) =: g \in K[Y]$, and hence $H(1-Y)\cdot b = g$. Write $H(1-Y) = \sum_b h_b(T)\,Y^b$ with the coefficients on the left. Taking coefficients gives

$$\sum_b h_b(t^n)\,\beta_{n-b} = g_n \qquad (n \ge 0), \tag{22}$$

a linear recurrence for β that is homogeneous for $n > \deg g$. In this normal order

$$(h(T)Y^b)(\kappa(T)Y^c) = h(T)\,\kappa(t^{-b}T)\,Y^{b+c},$$

so the unknown coefficients of H and R enter linearly. Thus an H of least Y-degree, which is the order of (22), is found by linear algebra over $K(T)$; one exists of degree $\le nD$ by the count of Lemma 4.1. This is the same computation as in Section 4.1: if $g = 0$, Proposition 4.4 decides whether $C \equiv 0$.

Sections. Fix a period p and a residue r, and put $x_m = \beta_{pm+r}$ and $\zeta = t^m$. Write (22) in companion form $v_{k+1} = A(t^k)v_k$, and put $B(\zeta) = A^{[p]}(t^r\zeta^p)$. The rows $w_0 = e_1$, $w_{j+1}(\zeta) = w_j(t\zeta)B(\zeta)$ satisfy $x_{m+j} = w_j(t^m)v_{pm+r}$. The first linear dependence among them over $K(\zeta)$ gives a proven recurrence

$$\sum_{i=0}^{k} q_i(t^m)\,x_{m+i} = 0 \qquad (m \ge m_0), \qquad q_i \in K[\zeta]. \tag{23}$$

Channels. A *q-hypergeometric term* is a sequence with $G(m+1) = g(t^m)\,G(m)$ for a rational $g \in K(\zeta)$, its *ratio*. The K-rational ratios of all q-hypergeometric solutions of (23) are computed by the q-analogue of Petkovšek's algorithm [5]. Write $g = Z \cdot A(\zeta)/B(\zeta) \cdot C(t\zeta)/C(\zeta)$ with $A \mid q_0$ and $B \mid q_k(t^{1-k}\zeta)$. The lowest and highest coefficients of the resulting equation determine Z and $\deg C$, and C is then found by linear algebra. Each ratio returned is verified by substitution.

Proposition 7.2 (decomposition)**.** *Let $q_k(t^m) \neq 0$ and $g_i(t^m) \notin \{0, \infty\}$ for $m \geq m_1 \geq m_0$, let g_i be ratios of solutions of* (23)*, and put $G_i(m) = \prod_{j=m_1}^{m-1} g_i(t^j)$. If $x_m = \sum_i a_i G_i(m)$ for the k values $m_1 \leq m < m_1 + k$, then this holds for all $m \geq m_1$.*

Proof. Both sides solve (23) for $m \geq m_1$. Since $q_k(t^m) \neq 0$, each value is determined by the k preceding ones. □

Proposition 7.3 (convergence of channels)**.** *Let $x_m = \sum_i a_i G_i(m)$ for $m \geq m_1$, with $\nu g_i(t^j) \geq 0$ for $j \geq m_1$ whenever $a_i \neq 0$. Call channel i* decaying *if $\nu g_i(t^j) \geq 1$ for all large j, and* stationary *otherwise. Suppose the stationary channels can be grouped into pairs (k_1, k_2) such that*

$$\theta = \frac{g_{k_2}}{g_{k_1}} = \theta_w(\zeta)\,\frac{\Gamma(t\zeta)}{\Gamma(\zeta)}, \qquad \Gamma \in K(\zeta), \quad \Gamma(0) = 1, \quad \Gamma(t^j) \notin \{0, \infty\}\ (j \geq m_1),$$

where θ_w satisfies (21) *for some $\lambda \geq 2$, offset μ with $\lambda m_1 + \mu \geq m_1$, and χ as in Proposition 7.1(ii), and*

$$a_{k_2}/a_{k_1} = -F(m_1)\,\Gamma(t^{m_1}), \qquad F(m) = \prod_{l=m}^{\lambda m+\mu-1} \chi(t^l).$$

Then $\nu(x_m) \geq sm + \kappa$ for explicit $s \geq 1$ and κ, and $S(\Delta) = 0$.

Proof. A decaying channel has $\nu(a_i G_i(m)) \geq \nu a_i + m - M_i$, as in Proposition 7.1(a). For a pair, (21) at $\zeta = t^m$ gives $F(m+1)/F(m) = \prod_i \chi(t^{\lambda m+\mu+i})/\chi(t^m) = \theta_w(t^m)$, so $G_{k_2}(m)/G_{k_1}(m) = \prod_{j=m_1}^{m-1} \theta(t^j) = F(m)\Gamma(t^m)/\big(F(m_1)\Gamma(t^{m_1})\big)$ and

$$a_{k_1} G_{k_1}(m) + a_{k_2} G_{k_2}(m) = a_{k_1} G_{k_1}(m)\big(1 - F(m)\,\Gamma(t^m)\big).$$

Every factor of $F(m)$ is some $\chi(t^l)$ with $l \geq m$, so $\nu(1 - F(m)) \geq i_1 m + c$ as in Proposition 7.1(b); and since $\Gamma(0) = 1$, $\nu(1 - \Gamma(t^m)) \geq \sigma m + \tau$ for large m, with $\sigma \geq 1$. This proves decay along one progression, and Corollary 3.4 concludes. □

Shifted factors. With $\Gamma = 1$ the pairing is that of Proposition 7.1(b). The factor Γ is needed when the two products differ by a finite boundary factor. For instance, $(-t;t)_{2m}$ and $1/(t^3;t^2)_m$ have $\theta = (1+t\zeta^2)(1+t^2\zeta^2)(1-t^3\zeta^2)$, which has no witness, but

$$(1-t)\,(-t;t)_{2m}\,(t^3;t^2)_m = (1-t^{2m+1}) \prod_{l=m+1}^{2m} (1-t^{2l}),$$

so $\Gamma(\zeta) = 1 - t\zeta^2$, $\theta_w = (1-t\zeta^2)(1+t\zeta^2)(1+t^2\zeta^2)$ and $\chi = 1 - t^2\zeta^2$ with $\lambda = 2$, $\mu = 0$. In general, θ_w is obtained by replacing each irreducible factor f of θ, normalized to $f(0) = 1$, by another member $f(t^s\zeta)$ of its class $\{f(t^s\zeta)\}_{s\in\mathbb{Z}}$; the implementation tries the members whose lowest coefficient has valuation in a window $[w, w+e)$, e the exponent of that coefficient, for $w = 0, \ldots, 3$, and then Γ is a product of shifts of the f.

Independence of the realization. Call two ratios g, g' *similar* if $g'/g = h(t\zeta)/h(\zeta)$ for some $h \in K(\zeta)^\times$. Terms with pairwise non-similar ratios are linearly independent over $K(\zeta)$; this is the q-analogue of the classical statement for hypergeometric terms [6]. Hence a sequence has at most one decomposition into q-hypergeometric terms with pairwise non-similar ratios. Moreover, if $x = \sum_i G_i$ satisfies a recurrence Q, then applying Q gives $\sum_i r_i G_i = 0$ with $r_i \in K(\zeta)$, so every $r_i = 0$: each constituent solves every recurrence of x. So when a section of β is a sum of q-hypergeometric terms, its constituents are found from the recurrence (23) of *any* realization. The outcome of Propositions 7.2 and 7.3 depends only on the sequence β, within the search bounds (the periods p tried, and the cap on $\deg C$). This route also subsumes the shifted certificates of Section 6 for the purpose of proving convergence.

Merging similar channels. The ratios returned by the q-Petkovšek step may be similar. Then the corresponding terms are proportional with rational factors, the coefficients a_i of Proposition 7.2 are not unique, and the individual terms can be stationary although their sum decays. So each similarity class is first replaced by one term: if $g_j = g_i\, h_j(t\zeta)/h_j(\zeta)$, then $G_j(m) = G_i(m)\, h_j(t^m)/h_j(t^{m_1})$ and

$$\sum_j a_j G_j(m) = P(t^m)\, G_i(m), \qquad P(\zeta) = \sum_j a_j \, \frac{h_j(\zeta)}{h_j(t^{m_1})},$$

a single term with ratio $g_i\, P(t\zeta)/P(\zeta)$; classes with $P = 0$ disappear. A solution h of $h(t\zeta)/h(\zeta) = \rho$ is found from the factorization of ρ. Its irreducible factors other than ζ fall into classes $\{f(t^s\zeta)\}_{s\in\mathbb{Z}}$; h exists iff ζ does not occur, the exponents in each class sum to zero and the remaining constant is a power t^k, and then $h = \zeta^k \prod f(t^s\zeta)^{-w_s}$, where w_s is the sum of the exponents of the class at shifts $\le s$.

7.5 Channels up to a convergent remainder

Proposition 7.3 needs a section of β that is a sum of q-hypergeometric terms. This fails for a *sum side*: a series such as $\sum_n t^{\binom{n+1}{2}}/(t;t)_n$, whose partial sums are not q-hypergeometric, while its product form $(-t;t)_\infty$ appears only in another realization (Section 9.6 has examples). Such a side is still a sum of q-hypergeometric terms up to a remainder that tends to 0, and the remainder is a combination of the sequence itself with coefficients regular at $\zeta = 0$.

Proposition 7.4 (remainders)**.** *Let $x_m \to S$ t-adically, and let $\pi_0, \dots, \pi_K \in K(\zeta)$ be regular at $\zeta = 0$ with $\sum_j \pi_j(0) = 0$. Then $\sum_j \pi_j(t^m)\, x_{m+j} \to 0$.*

Proof. The sum is $\sum_j \pi_j(t^m)(x_{m+j} - S) + S \sum_j \pi_j(t^m)$. The valuations of the $\pi_j(t^m)$ are bounded below, and $\sum_j \pi_j(t^m) \to \sum_j \pi_j(0) = 0$. □

This applies to $x_m = \beta^{(b)}_{pm+r}$, the partial sums of one diagonal block b of the system: $x_m \to S^{(b)}$ by Proposition 3.3, whatever the value $S^{(b)}$ is.

Hypergeometric elements. Let x solve the section recurrence (23),

$$(\mathcal{Q}x)_m = \sum_{i=0}^{k} q_i(t^m)\, x_{m+i} = 0 \qquad (m \ge m_0).$$

A *hypergeometric element* is a combination $w_m = \sum_{j=1}^{k} \ell_j(t^m) x_{m+j}$ with $\ell_j \in K(\zeta)$ that is a q-hypergeometric term. These are dual to the q-hypergeometric solutions of the adjoint recurrence

$$(\mathcal{Q}^* y)_m = \sum_{i=0}^{k} q_i(t^{m-i})\, y_{m-i} = 0.$$

Indeed, the concomitant

$$W_m = \sum_{i=1}^{k} \sum_{l=m-i+1}^{m} q_i(t^l)\, y_l\, x_{l+i}$$

satisfies $W_m - W_{m-1} = y_m(\mathcal{Q}x)_m - x_m(\mathcal{Q}^*y)_m$, so it is constant once both recurrences hold. If $y_{m+1} = \gamma(t^m)y_m$, then $w_m = W_m/y_m$ has ratio $1/\gamma$, and

$$\ell_j(\zeta) = \sum_{i=j}^{k} q_i(t^{j-i}\zeta) \prod_{u=1}^{i-j} \gamma(t^{-u}\zeta)^{-1}.$$

The ratios γ are found by the q-Petkovšek step applied to $\mathcal{Q}^*$, whose extreme coefficients are those of $\mathcal{Q}$, shifted. (Equivalently, the elements correspond to the first-order left factors of $\mathcal{Q}$, as the solutions correspond to its first-order right factors.)

The fit. Given hypergeometric elements $w_i = L_i x$ and Laurent polynomials $s_i, \rho \in K[\zeta, \zeta^{-1}]$, put $\Pi = 1 - \sum_i s_i L_i + \rho\, \mathcal{Q} = \sum_{j=0}^{k} \pi_j E^j$, with E the shift. Then

$$x_m = \sum_i s_i(t^m)\, w_{i,m} + (\Pi x)_m \qquad (m \geq m_0).$$

If every π_j is regular at $\zeta = 0$ and $\sum_j \pi_j(0) = 0$, the remainder $(\Pi x)_m$ tends to 0 (Proposition 7.4), and the terms $s_i w_i$ are channels whose sum has the same limit as x. These conditions are linear in the Laurent coefficients of the s_i and ρ: for a pole order e, the coefficients of $\zeta^{-e}, \dots, \zeta^{-1}$ in each π_j vanish, and the constant terms add up to 0. The implementation lets e run from 0 to 6, and then drops elements one at a time while the conditions stay solvable, so that few channels remain. Since a fit is only used to write x as channels plus a remainder, any solution is valid; the choice affects only whether the channels then pass Proposition 7.3.

The test with remainders. S is additive over the diagonal blocks. For each block, the section of its own β is written as channels, exactly (Proposition 7.2) or with a remainder. A fit found for $p = 1$ is restricted to the progression: $G(pm + r)$ has the ratio $\prod_{l<p} g(t^{r+l}\zeta^p)$ in m. If the union of the channels of all blocks, with similar classes merged, satisfies Proposition 7.3, then β_{pm+r} minus a sum of remainders tends to 0, so $S(\Delta) = 0$. The implementation tries this for each period after the plain channel test has failed for all combinations of readings.

Example 7.5. The first list of the second set in Section 9.6 contains the automata `[0R1 1U2 2R3 2U4 3R2 3U4 3U∞ 4U3]` and `[0R1 1U2 2R3 2U4 3R2 3U4 3U∞ 4R3]`, numbers 7 and 8, with the growth series $t\big((-t^2;t)_\infty - 1\big)$. In reading `X`, the β of number 7 is a sum of channels along even n: $\beta_{2m} = t/\big((1-t^2)(t^3;t^2)_{m-1}\big) - t$ for $m \geq 1$. No reading of number 8 has this property. In reading `Y` its β satisfies, with $\zeta = t^n$,

$$(t^2 + t^4\zeta)\beta_n - \big(t^2 - (t^3 - t^4)\zeta\big)\beta_{n+1} - (1 + t^3\zeta)\beta_{n+2} + \beta_{n+3} = 0,$$

whose only q-hypergeometric solutions are 1 and $(-t)^n$. The adjoint has the solutions with ratios $1/t$ and $1/(t + t^4\zeta)$, which give the elements

$$w_n^{(1)} = -(t + t^{n+4})\beta_{n+1} + (t - 1 - t^{n+3})\beta_{n+2} + \beta_{n+3}, \qquad w_n^{(2)} = -t\beta_{n+1} + (t-1)\beta_{n+2} + \beta_{n+3},$$

with ratios t and $t + t^{n+4}$. In both, the coefficients add up to 0 at $\zeta = 0$, so both tend to 0; the fit needs a pole, $s_1 = -s_2 = -1/((t^3 + t^4)\zeta)$ and $\rho = 0$, and then $\Pi = 1 - (tE + E^2)/(1 + t)$. Thus

$$\frac{t\beta_{n+1} + \beta_{n+2}}{1+t} = s_1(t^n)\, w_n^{(1)} + s_2(t^n)\, w_n^{(2)} = t\big((-t^2;t)_{n+1} - 1\big) \qquad (n \geq 0),$$

two channels with ratios 1 and $1 + t^{n+3}$, although β_n itself is not a sum of channels; the remainder $\beta_n - (t\beta_{n+1} + \beta_{n+2})/(1+t)$ has valuation $2\lfloor n/2 \rfloor + 3$. Restricted to even n, the constants cancel, and $t(-t^2;t)_{2m+1}$ pairs with the channel of number 7 through Euler's witness $\chi = 1 - t^2\zeta^2$, $\lambda = 2$, with a shifted form ($\Gamma = 1 + t\zeta^2 + t^2\zeta^2 + t^3\zeta^4$); this proves the difference.

8 Euler's odd–distinct identity

8.1 The input

Let

$$h = 1 + Y(1 - X - Y^2)^{-1}X, \qquad h' = 1 + Y(1 - Y - XY)^{-1}X, \qquad \Delta = h - h'.$$

The series $h - 1$ is the sum of all words $Y^{2i_1+1}XY^{2i_2}X \cdots Y^{2i_k}X$ with $k \geq 1$. Moving each X to the left costs one factor t per Y it passes, so this word equals $t^{|\lambda|}X^kY^{2i_1+\cdots+2i_k+1}$ with $\lambda = (2i_1 + 1,\ 2i_1 + 2i_2 + 1,\ \ldots,\ 2i_1 + \cdots + 2i_k + 1)$, the partition into odd parts read off from the number of Y's preceding each X. Similarly the words of $h' - 1$, $Y\,w\,X$ with $w \in \{Y, XY\}^*$, encode the partitions into distinct parts.

These expressions are not normalized realizations (3): $1 - X - Y^2 \not\equiv 1 \pmod t$, and indeed $(1 - X - Y^2)^{-1}$ does not converge t-adically, since all the words X^k have weight t^0. Only the outer letters $Y \cdots X$ produce the weight. To move them inside, let σ be the automorphism $X \mapsto tX$, $Y \mapsto tY$ of A. For a normal monomial $m = X^aY^b$, both YmX and $tX\sigma(m)Y$ equal $t^{a+b+1}X^{a+1}Y^{b+1}$; hence $YwX = tX\sigma(w)Y$ for every word w. Put $P_{\mathrm{o}} = \sigma(1 - X - Y^2) = 1 - tX - t^2Y^2$ and $P_{\mathrm{d}} = \sigma(1 - Y - XY) = 1 - tY - t^2XY$, both $\equiv 1 \pmod t$. As w runs over $\{X, Y^2\}^*$, the words $\sigma(w)$ are the terms of the convergent expansion of P_{o}^{-1}, and likewise for $\{Y, XY\}^*$ and P_{d}^{-1}. Summing over words gives

$$h = 1 + tXP_{\mathrm{o}}^{-1}Y, \qquad h' = 1 + tXP_{\mathrm{d}}^{-1}Y.$$

A normalized realization of Δ is

$$Q = \mathrm{diag}(P_{\mathrm{o}}, P_{\mathrm{d}}), \qquad r = (X, X), \qquad e = (tY, -tY)^{\mathsf{T}},$$

with compiled triple

$$L = \begin{pmatrix} 1 - tT - t^2Y^2 & 0 \\ 0 & 1 - tY - t^2YT \end{pmatrix}, \qquad f = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad d = (tY, -tY). \tag{24}$$

Writing $m_{\mathrm{o}} = \sum \eta_n Y^n$ and $m_{\mathrm{d}} = \sum \zeta_n Y^n$, coefficient extraction gives $\eta_{2j+1} = 0$ and

$$\eta_{2m} = \frac{t^{2m}}{(t; t^2)_{m+1}}, \qquad \zeta_n = t^n(-t; t)_n, \qquad S(\Delta) = t\sum_{n \geq 0}(\eta_n - \zeta_n).$$

We will see that $\beta_{2m} = (t; t^2)_m^{-1} - (-t; t)_{2m}$, so $S(\Delta) = (t; t^2)_\infty^{-1} - (-t; t)_\infty$: the identity $S(\Delta) = 0$ is Euler's theorem that partitions into odd parts and into distinct parts are equinumerous [3, 26.10].

8.2 The identity: one inverse shift

Equations (13), with the condition $(R \cdot f)(1) = 0$, read

$$(1 - tz)R_1(z) - t^2R_1(t^2z) = t, \qquad R_2(z) - t(1 + tz)R_2(tz) = -t, \qquad R_1(1) + R_2(1) = 0.$$

(the last condition makes $\kappa = 0$ in Proposition 5.2). There is no polynomial solution: $-tzR_1(z)$ would be the unique top-degree term. The homogeneous equations $(1 - tz)h(z) = t^2h(t^2z)$ and $h(z) = t(1 + tz)h(tz)$ have no rational solution, since the divisor of $h(t^2z)/h(z)$, resp. $h(tz)/h(z)$, has degree 0 on every orbit $\alpha t^{\mathbb{Z}}$ while $1 - tz$ and $1 + tz$ have a single zero. So the rational solution

is unique, $R = (-T^{-1}, T^{-1})$, with a pole of order $s = 1$ at $T = 0$, as Proposition 6.2 predicts. Left division by $1 - Y$ gives $P = ((1 + Y)T^{-1} - t,\ -T^{-1})$. In the shifted form (16),

$$\widetilde{L} = \begin{pmatrix} 1 - Y^2 - tT & 0 \\ 0 & 1 - Y - tYT \end{pmatrix}, \quad \widetilde{f} = f, \quad \widetilde{d} = (tYT, -tYT),$$

$$\widetilde{d} = (1 - Y)P_s + R_s\widetilde{L}, \qquad P_s = (1 + Y - tT,\ -1), \quad R_s = (-1, 1), \quad R_s\widetilde{f} = 0, \tag{25}$$

(e.g. $(1 - Y)(1 + Y - tT) - (1 - Y^2 - tT) = tYT$). Hence, with $\mu = T^{-1}m$,

$$b = (1 + Y - tT)\mu_{\mathrm{o}} - \mu_{\mathrm{d}}, \qquad (1 - Y)b = C. \tag{26}$$

The auxiliary series are integral but divergent: modulo t they are $(1 - Y^2)^{-1}$ and $(1 - Y)^{-1}$. Only the output b will be shown to converge.

8.3 Convergence: channels and a dilation witness

Write $\mu_{\mathrm{o}} = \sum x_n Y^n$, $\mu_{\mathrm{d}} = \sum y_n Y^n$. From $\widetilde{L}\mu = f$,

$$(1 - t^{n+1})x_n = x_{n-2} + \delta_{n0}, \qquad y_0 = 1, \quad y_n = (1 + t^n)y_{n-1}.$$

The state $v_n = (x_n, x_{n-1}, y_n)$ satisfies $v_{n+1} = A(t^n)v_n$, and (26) gives $\beta_n = \pi(t^n)v_n$ with

$$A(z) = \begin{pmatrix} 0 & \frac{1}{1-t^2z} & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 + tz \end{pmatrix}, \qquad \pi(z) = (1 - tz,\ 1,\ -1).$$

The two-step product $A(tz)A(z)$ is diagonal. Along $n = 2m$ the middle channel vanishes ($x_{\mathrm{odd}} = 0$), and the others are

$$U_m = (1 - t^{2m+1})x_{2m} = \frac{1}{(t; t^2)_m}, \qquad V_m = y_{2m} = (-t; t)_{2m}, \qquad \beta_{2m} = U_m - V_m,$$

with multipliers $g_U(\zeta) = (1 - t\zeta^2)^{-1}$ and $g_V(\zeta) = (1 + t\zeta^2)(1 + t^2\zeta^2)$, $U_0 = V_0 = 1$, so the initial ratio is -1. The ratio is $\theta(\zeta) = (1 - t\zeta^2)(1 + t\zeta^2)(1 + t^2\zeta^2)$; for $\lambda = 2$ the degree equation gives $\deg\chi = 6/3 = 2$, and the recursion returns

$$\chi(\zeta) = 1 - t^2\zeta^2, \qquad \chi(\zeta^2)\chi(t\zeta^2) = (1 - t^2\zeta^4)(1 - t^4\zeta^4) = \theta(\zeta)\chi(\zeta). \tag{27}$$

Proposition 7.1(b) applies with $i_1 = c = 2$:

$$\beta_{2m} = U_m\Big(1 - \prod_{j=m}^{2m-1}(1 - t^{2j+2})\Big), \qquad \nu(\beta_{2m}) \geq 2m + 2.$$

The same channels, witness and bound are obtained from the scalar recurrence of β (Section 7.4), whichever realization of the two series is used: PBW expressions, automata, or redundant automata. By Corollary 3.4, $S(\Delta) = 0$, $b \in B^{\mathrm{hol}} \cap \widehat{B}$, and $\nu(\beta_n) \geq k_\Delta(n) = \lceil n/2 \rceil$ for all n (here $\alpha = 1$, $D = 2$). Proposition 3.2 then supplies the unique $a \in \widehat{A}$ with $\Delta = (1 - X)a + b(1 - Y)$. The odd indices need no separate argument.

Remark 8.1 (other routes). The row π can also be found directly as a vector telescoper from the unshifted system (24): the state $v_n = (\eta_n, \zeta_n, \eta_{n-1})$ satisfies $v_{n+1} = A_0(t^n)v_n$ with

$$A_0(z) = \begin{pmatrix} 0 & 0 & \frac{t^2}{1-t^2z} \\ 0 & t(1 + tz) & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad c_{n+1} = (t, -t, 0)\, v_n,$$

and the row $\pi_0(z) = \big((1-tz)/z,\, -1/z,\, t/z\big)$ satisfies $\pi_0(tz)A_0(z) - \pi_0(z) = (t, -t, 0)$ and $\pi_0(1)v_0 = 0$, so $\beta_n = \pi_0(t^n)v_n$ without any operator certificate. Uniform contraction cannot prove its decay: for the initial vector $((1-t)^{-1}, 0, 0)$ the same recurrence and row yield the unit sequence U_m at even indices, so no bound valid for all initial vectors exists. The decay is a property of the selected initial values, which is exactly what condition (i) of Proposition 7.1 records.

9 Readings: other series with the same sum

The test of Section 7.4 depends only on the partial sums β of the marginal C. The marginal, however, depends on the realization, while $S(\Delta) = C(1)$ does not. Partition generating functions typically have two marginals of different nature. For partitions into distinct parts ≥ 2, with sum $S = (-t^2; t)_\infty - 1$,

$$C_{\mathrm{lp}} = \sum_{n\geq 2} t^n (-t^2; t)_{n-2}\, Y^n, \qquad C_{\mathrm{np}} = \sum_{k\geq 1} \frac{t^{k(k+3)/2}}{(t;t)_k}\, Y^k$$

count them by largest part and by number of parts. The partial sums of C_{lp} are products, $\beta_n = (-t^2; t)_{n-1} - 1$, with $\nu(S - \beta_n) = n+1$. Those of C_{np} are partial sums of a q-hypergeometric series, with $\nu(S - \beta_k) = (k+1)(k+4)/2$. When the two sides of Δ are realized in these two ways, $C \not\equiv 0$ and the channel test finds no proof (systems 1 and 2 of Section 9.4 are an example), although $S(\Delta) = 0$. This section replaces each block of the marginal system by other systems with the same sum.

9.1 The second marginal

Let ρ_Y be evaluation at $Y = 1$, followed by renaming X to Y, and let $\Psi : A \to \mathcal{O}$ be the isomorphism $\Psi(X^a Y^b) = Y^a T^b$. On normal monomials of $\mathcal{O}$ put $\mathrm{sw}(Y^b T^a) = Y^a T^b$, extended linearly.

Lemma 9.1 (transposition)**.** *Let (L, f, d) be a normalized marginal system with sum S. Then*

$$L'_{ij} = \mathrm{sw}(L_{ji}), \qquad f'_i = \mathrm{sw}(d_i)\big|_{T=1}, \qquad d'_j = f_j(T)$$

is a normalized marginal system with the same sum S.

Proof. Put $Q = \Phi^{-1}(L^{\mathsf{T}})$, $e = \Phi^{-1}(d)$ and $r_j = f_j(Y) \in A$. By (4), (L, f, d) is the marginal system of the right realization $wQ = r$, $\Delta = we$, so $S(\Delta) = S$. The same element is $\Delta = rw'$ with $Qw' = e$, a *left* realization, and $w' = Q^{-1}e$ is again given by geometric inversion. For $q = X^a Y^b$ and $w = X^i Y^j$ we have $qw = t^{bi} X^{a+i} Y^{b+j}$, hence $\rho_Y(qw) = \Psi(q) \cdot \rho_Y(w)$. Applying ρ_Y to $Qw' = e$ and to $\Delta = rw'$ gives $L' \cdot m' = f'$ and $\rho_Y(\Delta) = d' \cdot m'$, with $m' = \rho_Y(w')$, $L' = \Psi(Q)$, $f' = \rho_Y(e)$ and $d' = \Psi(r)$; these are the formulas above. Both marginals of Δ take the value $S(\Delta)$ at 1. □

In the encoding of Section 8 a partition λ becomes $t^{|\lambda|} X^k Y^j$ with k parts and largest part j. So the two marginals of such a realization count by largest part and by number of parts.

9.2 Canonical realizations

The second marginal depends on the system, not only on C. A system determined by C comes from an annihilator. Let $H \cdot C = g \in K[Y]$ as in Section 7.4, write $H = \sum_b h_b(T)\, Y^b$ with coefficients on the left, and let $h \in K[T]$ be the greatest common divisor of the h_b, the *content*: $H = h(T)H'$.

Proposition 9.2 (canonical realization)**.** *The series $u = H' \cdot C$ is a polynomial: $h(t^n)u_n = g_n$, so $u_n = 0$ for $n > \deg g$ except at the finitely many n with $h(t^n) = 0$, and u is computed from finitely many c_n. If $t^{-v}H' \equiv c \pmod t$ for some $v \in \mathbb{Z}$ and $c \in \mathbb{Q}^\times$, then*

$$\big(t^{-v}H'/c, \quad t^{-v}u/c, \quad 1\big)$$

is a normalized 1×1 marginal system whose marginal is C. If H has least Y-degree among all H with $H \cdot C \in K[Y]$, the transpose of this system (Lemma 9.1) depends only on C.

Proof. $h(T)$ acts on Y^n as multiplication by $h(t^n)$, and the zeros of h at $z = t^n$ are bounded as in Section 7. The system is normalized and $C \in \widehat{B}$ solves it, so C is its selected solution. The H with $H \cdot C \in K[Y]$ form a left ideal of $\mathcal{O}$. If H_1, H_2 are elements of least degree, a combination $b(T)H_1 - a(T)H_2$ of lower degree lies in the ideal, so it vanishes: H_1 and H_2 are proportional over $K(T)$, and their content-free parts differ by a factor $\lambda \in K^\times$. Replacing (H', u) by $(\lambda H', \lambda u)$ multiplies L' and d' of the transpose by λ and leaves its marginal $d' \cdot (L')^{-1} \cdot 1$ unchanged. □

The implementation takes H from `annihilator`, of least degree under a common bound on the degrees of H and R. Only the last statement of the proposition depends on minimality; every reading has the sum S in any case.

9.3 Searching the readings

If $\Delta = h - h'$ comes from two realizations, its marginal system is block diagonal; in general the blocks are the connected components of the nonzero pattern of L. The sum is additive over blocks, so every block can be replaced by any of its *readings*: the block itself (`Y`), its transpose (`X`), and the transposes of the canonical realizations of these two (`cY′`, `cX′`), when they exist. Every combination is a normalized system with the same sum, and all previous methods apply to it unchanged: a violated a-priori rate proves $S(\Delta) \neq 0$, and a zero certificate, a rational certificate without pole at $T = 0$, a polynomial certificate or a channel proof proves $S(\Delta) = 0$. The annihilator of a combination is assembled from those of its blocks: if $H_k \cdot C_k = g_k$ and $H = A_1H_1 = A_2H_2 = \cdots$ is a least common left multiple, then $H \cdot \sum_k C_k = \sum_k A_k g_k$. Readings of one series may also differ by a power of Y: the second marginal of a realization counts the parts from 0 or from 1, depending on where the outer letters are placed. Since $(Y^s C_k)(1) = C_k(1)$, when the marginals of the blocks start at different powers of Y and cancel on their first coefficients after multiplying each block by the power of Y that aligns them ($d \mapsto Y^s d$), the aligned combination is tried as well, and the zero test decides it. The implementation runs the cheap tests on the system as given, then on all combinations (at most four readings per block), and then the channel test with increasing period across all combinations.

9.4 Six realizations of one series

The following six systems all have the sum $S = (-t^2; t)_\infty - 1 = t^2 + t^3 + t^4 + 2t^5 + 2t^6 + 3t^7 + \cdots$ of partitions into distinct parts ≥ 2; in each, $f = e_1$.

1 $L = \begin{pmatrix} 1 - tT & -tT \\ -tY & 1 \end{pmatrix}$, $d = (0, tY)$

2 $L = \begin{pmatrix} 1 & -tT \\ -tY & 1 - tY \end{pmatrix}$, $d = (0, tY)$

3 $L = \begin{pmatrix} 1 & -tY \\ -tT & 1 - tT \end{pmatrix}$, $d = (0, tY)$

4 $L = \begin{pmatrix} 1 & -tY & 0 \\ -tY & 1 & -tY \\ -tT & -tT & 1 \end{pmatrix}$, $d = (0, tY, 0)$

5 $L = \begin{pmatrix} 1 & -tY & 0 \\ -tY & 1 & -tT \\ -tT & -tT & 1 \end{pmatrix}$, $d = (0, tY, 0)$

6 $L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 - tY & -tY \\ -tT & -tT & 1 \end{pmatrix}$, $d = (0, tY, tY)$

Their readings are of four kinds, distinguished by the valuations $\nu(c_n)$: by largest part (lp, $\nu(c_n)$ linear in n), by number of parts (np, quadratic), with alternating parity (alt: $2, 5, 4, 7, 6, 9, \dots$), and supported on even n (ev):

reading	1	2	3	4	5	6
Y	np	lp	np	alt	alt	lp
X	lp	np	lp	quadratic	ev	np
cY′	–	np	–	np	–	np
cX′	np	lp	np	lp	lp	–

(–: no canonical realization, since $t^{-v}H' \not\equiv c \pmod t$.) All fifteen differences $\Delta_i - \Delta_j$ are proved, with the readings of Δ_i and Δ_j shown: by a rational certificate without pole at $T = 0$ for the least annihilator (R), by the zero test (Z: the two readings are equal series), or by the channel test along $n = 2m + r$ (C_r):

	2	3	4	5	6
1	R X,Y	R Y,Y	R X,Y	C_0 X,X	R X,Y
2		R X,Y	R Y,Y	C_0 Y,X	R Y,Y
3			R X,Y	C_0 X,X	Z X,Y
4				C_0 cX′,X	R Y,Y
5					C_0 X,Y

In the nine R cases the two readings are different series, and their difference has a least annihilator of order 1 or 2, against 2 or 3 for the annihilator of the system; for 1–3, for instance, $R = (t^2 + t)/\big((t + 1)^2T^2 - (t^2 + 2t)T + t\big)$.

In the five channel proofs, β_{2m} is a combination of two product channels with ratios $1/(1 - t^3\zeta^2)$ and $(1+t^a\zeta^2)(1+t^{a+1}\zeta^2)$, whose quotient is $\theta(\zeta) = (1-t^3\zeta^2)(1+t^a\zeta^2)(1+t^{a+1}\zeta^2)$. For $a \in \{2, 3\}$ ($a = 3$ for 1–5 and 4–5, $a = 2$ for 3–5 and 5–6) the witness is $\chi = 1 - t^{2a-2}\zeta^2$ with $\lambda = 2$ and offset $\mu = 1$:

$$\chi(t\zeta^2)\,\chi(t^2\zeta^2) = (1 - t^{2a}\zeta^4)(1 - t^{2a+2}\zeta^4) = \theta(\zeta)\,\chi(\zeta).$$

For $a = 1$ (pair 2–5) there is no witness for θ itself, and the proof uses the shifted form of Section 7.4: $\Gamma = 1 - t\zeta^2$ and $\chi = 1 - t^2\zeta^2$ with $\mu = 0$. With the systems as given (the readings Y alone), only the four differences 1–3, 2–4, 2–6 and 4–6 are proved.

9.5 Four families

Three more families were treated in the same way. Together with the six systems above (family A) there are 27 systems in four families, with one sum per family:

$$S_A = (-t^2;t)_\infty - 1, \qquad S_B = (-t;t)_\infty - 1 - t, \qquad S_C = t\,(-t;t)_\infty, \qquad S_D = t\,\big((-t;t)_\infty - 1\big).$$

In all of them $f = e_1$; the matrices L are written row by row, as in the input of the implementation:

B1 $[1,0,0;\ -tT, 1-tT, -tY;\ -tY, -tY, 1]$ $(0, tY, 0)$
B2 $[1,0,0;\ -tY, 1-tY, -tY;\ -tT, -tT, 1]$ $(0, tY, 0)$
B3 $[1,0,0;\ -tY, 1-tY, -tT;\ -tT, -tT, 1]$ $(0, tY, 0)$
B4 $[1-tT, -tT, 0;\ -tY, 1, 0;\ 0, -tY, 1]$ $(0, tY, tY)$
B5 $[1,0,0;\ -tT, 1-tY, -tY;\ 0, -tT, 1]$ $(0, tY, tY)$
B6 $[1,0,0;\ 0, 1, -tY;\ -tT, -tT, 1-tT]$ $(0, tY, tY)$

C1 $[1-tY, -tY, 0;\ 0, 1, -tY;\ -tT, -tT, 1]$ $(tY, 0, 0)$
C2 $[1-tY, 0, -tY;\ -tT, 1, -tT;\ 0, -tT, 1]$ $(tY, 0, 0)$
C3 $[1,0,0;\ -tT, 1-tT, -tY;\ 0, -tY, 1]$ $(tY, tY, 0)$
C4 $[1,0,0;\ -tT, 1-tY, -tY;\ 0, -tT, 1]$ $(tY, tY, 0)$
C5 $[1,0,0;\ -tT, 1-tY, -tT;\ 0, -tT, 1]$ $(tY, tY, 0)$
C6 $[1,0,-tY;\ -tY, 1, 0;\ -tT, 0, 1-tT]$ $(tY, tY, 0)$
C7 $[1,0,0;\ -tY, 1, -tY;\ -tT, -tY, 1-tT]$ $(tY, tY, 0)$
C8 $[1,0,0;\ -tT, 1, -tY;\ 0, -tT, 1-tT]$ (tY, tY, tY)

D1 $[1-tT, -tY;\ -tY, 1]$ $(0, tY)$
D2 $[1, -tT;\ -tT, 1-tY]$ $(0, tY)$
D3 $[1,0,0;\ -tT, 1-tT, -tY;\ 0, -tY, 1]$ $(0, tY, 0)$
D4 $[1,0,0;\ -tT, 1-tY, -tY;\ 0, -tT, 1]$ $(0, tY, 0)$
D5 $[1,0,-tY;\ -tY, 1-tY, 0;\ -tT, -tT, 1]$ $(0, tY, 0)$
D6 $[1,0,-tT;\ -tY, 1-tY, 0;\ -tT, -tT, 1]$ $(0, tY, 0)$
D7 $[1,0,0;\ -tT, 1, -tY;\ 0, -tT, 1-tT]$ $(0, tY, tY)$

All 79 differences within a family are proved:

family	sum	systems	differences	zero test	rational	channels
A	$(-t^2;t)_\infty - 1$	6	15	1	9	5
B	$(-t;t)_\infty - 1 - t$	6	15	1	6	8
C	$t\,(-t;t)_\infty$	8	28	0	12	16
D	$t\,((-t;t)_\infty - 1)$	7	21	2	7	12
all		27	79	4	34	41

The 41 channel proofs are product identities in the sense of Proposition 6.2. In the readings used, no annihilator of lower order than that of the system (order 3) is found, the least pole at $T = 0$ of a rational certificate has order 2, and β_{2m} consists of two product channels that cancel through one dilation witness, $\chi = 1 - t^2\zeta^2$ (38 pairs) or $1 - t^4\zeta^2$ (3), with $\lambda = 2$ and offset $\mu = 0$ (33) or $\mu = 1$ (8); 20 of them need a shifted form ($\Gamma \neq 1$). In the other 38, the readings found first, all of them `X` or `Y`, are equal series (zero test, 4 pairs), differ by a nonzero polynomial (2 pairs, $A_C = 1$), or differ by a series whose least annihilator has order 1 (23 pairs) or 2 (9 pairs); the annihilator of the system has order up to 4. Every rational certificate found is without pole at $T = 0$. For B1–B3, for instance, the readings $(\mathtt{X}, \mathtt{Y})$ differ by a q-hypergeometric series, the system gives an annihilator of order 2, and the certificate for A_C is

$$R = \frac{t^3}{(t^2T^2 + t - 1)(T^2 + t - 1)}, \qquad R(0) = \frac{t^3}{(t-1)^2}.$$

A canonical reading occurs in one proof (A4–A5). With the systems as given, without other readings, 23 of the 79 differences are proved.

9.6 Lists of automata with a common growth series

A larger test consists of 497 lists of automata over two letters, with conjecturally the same growth series within each list. An automaton is given by its transitions iVj, from state i to state j reading $V \in \{R, U\}$, with initial state 0 and final state ∞; for instance `[0R1 1R2 1U1 1U∞ 2R1]` recognizes $R(U + R^2)^*U$. With $R = Y$ and $U = X$, an automaton stands for the sum of its accepted words, and its growth series is S of that sum: the generating function of the words by the number of pairs $R \cdots U$ (Euler's odd series in the example). Every accepted word has the form $R \cdots U$, so the sum is $Y(\sum_w w)X$ over the inner words, realized as in Section 8 (`automaton` reads this notation). The lists contain 1116 automata with two or three inner states, in 422 pairs and 75 lists of three to eight; within each list the series agree up to t^{40}. That makes 813 differences.

All 813 differences are proved directly (`automata_quantumwz.jl`), with a time limit of 300 seconds for the channel search:

rational certificate without pole at $T = 0$	728
zero test (two after aligning the readings by a factor Y)	14
channel test, period 2, $\lambda = 2$	69
channel test, period 3, $\lambda = 3$	2
all	813

Most proofs use the readings $(\mathtt{X}, \mathtt{Y})$ (485) or $(\mathtt{Y}, \mathtt{Y})$ (296); canonical readings occur in five channel proofs. Of the 71 channel proofs, 39 need a shifted form θ_w ($\Gamma \neq 1$), as for the boundary factor in Euler's identity of Section 7.4; the witnesses are $1 - t^2\zeta^2$ (66), $1 - t^4\zeta^2$ (3) and $1 - t^3\zeta^3$ (2,

with period 3). The median time per difference is 0.2 seconds; 11 take more than ten seconds, the longest two and a half minutes, and all 813 take 17 minutes. Three changes to the method came from this data: the least annihilator is reconstructed from modular images (the direct solution over K took minutes for systems of 20 unknowns), stationary channels may differ by a boundary factor (the shifted forms θ_w), and readings may differ by a power of Y (the aligned combinations).

A second set. A second file has 49 lists of 10 to 19 automata with two to four inner states: 579 automata and 3237 differences. Run with the remainder test of Section 7.5 and a time limit of 150 seconds, all but 29 differences are proved directly:

rational certificate without pole at $T = 0$	2726
zero test	70
channel test, period 2, $\lambda = 2$	399
channel test with a remainder, period 2, $\lambda = 2$	13
not proved (Gauss's identity, below)	29
all	3237

The remainder test was added for this set. In the first list, automaton 8 is the sum side of Example 7.5; its differences with numbers 7 and 10 were not proved before, and the list was certified only through the other differences. All 13 differences of this kind, in four lists, are now proved, each with a fit of pole order 1 for one block, found for $p = 1$ and restricted to a progression of period 2. Of the 412 channel proofs, 273 need a shifted form; the witnesses are $1 - t^2\zeta^2$ (401) and $1 - t^4\zeta^2$ (11). Most proofs use the readings $(\mathtt{Y}, \mathtt{Y})$ (2485) or $(\mathtt{X}, \mathtt{Y})$ (548); canonical readings occur in 10. The median time of a proof is 0.1 seconds and the longest 28 seconds; the proved differences take 44 minutes in all, and the 29 others 94 minutes, until the time limit (one step of the search can overrun it by a few minutes). So 47 of the 49 lists are certified.

Gauss's identity. The 29 unproved differences are those of automaton 10 of list 34, and of automata 6 and 7 of list 42, with the other members of their lists (the difference of 6 and 7 is proved, by the zero test). Automaton 10 of list 34 is `[0R1 1R2 1U3 1U∞ 2R1 2U2 3R4 3U4 3U∞ 4R3]`. Its words are $R\,w_1 \cdots w_k\,U$ and $R\,w_1 \cdots w_k\,U\,v_1 \cdots v_l\,U$ with factors $w_i = RU^{a_i}R$ and $v_j \in \{UR, RR\}$. The letters U of w_i follow $2i$ letters R, so the factors w_i contribute $1/(t^2;t^2)_k$, and $N = 2k + 1$ letters R precede the next U. An ending with u factors UR and r factors RR has the total weight $t^{N(u+2)+u(u+1)/2+2r}\binom{u+r}{u}_{t^2}$. Summing over r, then over k by $\sum_k z^k/(t^2;t^2)_k = 1/(z;t^2)_\infty$,

$$S = \frac{t}{(t^2;t^2)_\infty}\Big(1 + \sum_{u\ge 0} t^{\binom{u+2}{2}}\Big) = \frac{t\,\psi(t)}{(t^2;t^2)_\infty}, \qquad \psi(t) = \sum_{v\ge 0} t^{\binom{v+1}{2}},$$

while the other members of the list have $S = t(-t;t)_\infty$. The same computation gives $S = \psi(t)/(t^2;t^2)_\infty - 1$ for automaton 6 of list 42, `[0R1 1R2 1U3 1U∞ 2R1 2U2 2U∞ 3R4 4R3 4U3 4U∞]`, whose list has $S = (-t;t)_\infty - 1$. These equalities are Gauss's identity $\psi(t) = (t^2;t^2)_\infty/(t;t^2)_\infty$, a theta-function identity. None of the three automata has a reading in which β is a sum of channels, exactly or up to a remainder, for $p \le 2$: in reading $\mathtt{X}$, the partial sums of automaton 10 have the q-hypergeometric constituents 1 and $1/(t^4;t^2)_n$ and a third part that is not a sum of channels, and the hypergeometric elements of their module give no fit (pole order up to 10). So lists 34 and 42 are the two lists that the tests do not certify. This is a limitation of the certificate classes: they prove identities between products, and sum sides that a finite identity within the same module relates to a product (as in Example 7.5), but not a theta series such as ψ.

10 Implementation

The accompanying module `QuantumWZ.jl` implements the route of Sections 7.4, 7.5 and 9. Both algebras are OSCAR PBW algebras [7] over $K = \mathbb{Q}(t)$: `A` with relation $YX = tXY$ and `O` with $TY = tYT$, with normal monomials X^aY^b and Y^bT^a as in the paper. Operators stay PBW elements throughout and are read by a single function, `pbw_terms`. A power series is stored as its vector of coefficients, and the action of $\mathcal{O}$ is used only through the formula $[Y^k](cY^bT^a \cdot v) = c\,t^{a(k-b)}v_{k-b}$ (`ycoeff`). Recurrences live in $F = \mathbb{Q}(t, z)$, the fraction field of the Flint polynomial ring $\mathbb{Q}[t, z]$, where z stands for t^n (or for T); factorization over K is factorization in $\mathbb{Q}[t, z]$. Linear algebra is plain Gauss–Jordan elimination on matrices of normalized fractions (`rref_field`, `kernel_field`, `solve_field`), over K whenever the entries are free of z; on the annihilator systems it is faster than the generic fraction-free routines by two to three orders of magnitude.

1. `compile` produces (L, f, d) by (4), and `difference` the block-diagonal system of $\Delta_1 - \Delta_2$ from two realizations or two compiled systems; `automaton` also reads the notation of Section 9.6. `attempt` splits the system into `blocks` and runs the following steps (`cheap_stages`) on the system as given; if nothing is decided, it lists the `readings` of each block (`transpose_system`, `canonical`) and runs them on every other combination, together with the aligned combinations of Section 9.3 (`aligned`).
2. `solve_series` and `marginal_coeffs` give $c_0, \dots, c_N$ and $\beta_0, \dots, \beta_N$; N is extended automatically when a proof needs more terms. `nonzero_witness` applies Proposition 3.3(1).
3. `annihilator` computes H and R with $Hd = RL$ over $K(T)$, of least degree under a common degree bound, and checks the identity in `O`. A block-diagonal system is done block by block, with H a least common left multiple of the block annihilators (`lclm`). If $g = \varepsilon(Rf) = 0$, `exceptional_set` and Proposition 4.4 decide $C \equiv 0$ (`:zero_marginal`).
4. `reduce_annihilator` looks for a right factor A of H of lower order with $A \cdot C = g_A$ a polynomial (Proposition 5.5). The coefficients of A, polynomials in T, are guessed from the c_n: the least order and then the least degree are found on modular images ($t \mapsto T_0$ in $\mathrm{GF}(2^{61} - 1)$, `marginal_mod`) with twice as many equations as unknowns. The kernel, one-dimensional, has entries in $\mathbb{Q}(t)$ but is computed from its images at further points $t \mapsto \tau$: each entry is interpolated and rationally reconstructed in t, then its coefficients are lifted to $\mathbb{Q}$ (`reconstruct_kernel`); only for small systems is it solved over K directly, since the c_n have large t-degree. The guess is then proved by `check_factor`: right division $H = QA$, the identity $\widetilde{Q}A = DH$ in `O`, $\widetilde{Q} \cdot g_A = D \cdot g$, and the exceptional indices of $\widetilde{Q}$. The search is limited to 32 unknowns; beyond it, or if the reconstruction fails, $A = H$.
5. `rational_certificate` solves (15) with A in place of H for rational R: `universal_denominator`, the pole bound at 0 and the degree bound, then the numerator by forward substitution in its coefficients (`polynomial_solution`: the coefficient of z^{v+i} determines the i-th unknown except where $\chi(t^i) = 0$, so only those few free parameters are found by linear algebra). Without a pole at $T = 0$ it returns `:rational`, or `:nonzero` with the exact value $S(\Delta) = \kappa \neq 0$. Inside `attempt` the search is skipped when the cleared denominator has degree above 120. `polynomial_certificate` looks for a polynomial certificate of the system itself (`:polynomial`).
6. For $p \leq p_{\max}$, every combination and every residue, `progression_test`
   - turns $A(1 - Y)$ into (22) (`beta_recurrence`) and its section (23) (`section_recurrence`);
   - finds the q-hypergeometric solutions (`qhyper`); a candidate is rejected from the extreme coefficients of the T_i alone, before any polynomial is formed;
   - fixes the coefficients a_i from k values (`decompose`, Proposition 7.2) and merges similar channels (`merge_similar`, `similarity`);
   - applies Proposition 7.3 (`channel_proof`), with dilation witnesses for every admissible λ and offsets $|\mu| \leq 6$ (`dilation_witness`), for θ itself and for its shifted forms θ_w (`shift_forms`).

Success gives `:channels`, with the progression, the channels, the witnesses and the bound $\nu(\beta_{pm+r}) \geq sm + \kappa$.

7. For each period, once `progression_test` has failed for every combination, `remainder_test` runs the test with remainders of Section 7.5. Each block gets its own least annihilator and β recurrence (`block_reading`); `block_channels` writes the section of its β as channels, exactly or with a remainder (`hyper_elements`: `qhyper` on the adjoint and the concomitant; `remainder_fit`: the Laurent conditions; `restrict`: a fit for $p = 1$ along the progression). The channels of all blocks are merged (`merge_channels`) and passed to `channel_proof`. Block data are cached across combinations. Success gives `:channels` with the field `remainders`: for each block, `:none` or the pole order of its fit.

Every result records the readings used. The result `:unknown` is not a disproof.

Tests. `test_quantumwz.jl` checks:

- $X - Y$ (`:rational`, no denominator), the second system of Example 5.6 (`:rational`, denominator $1 + T^2$), and $\Delta = 1$ (`:nonzero` at $n = 0$);
- Example 5.6 and Example 4.6 (`:zero_marginal`), together with further pairs of equal series given by different realizations (PBW, automata, a redundant automaton);
- Euler's identity for four pairs of realizations (PBW and PBW, automaton and automaton, PBW and automaton, automaton and redundant automaton) and for $-\Delta$. All give the proof of Section 8: readings $(\mathtt{Y}, \mathtt{Y})$, period 2, a single dilation pair with $\lambda = 2$ and $\chi = 1 - t^2 z^2$, and $\nu(\beta_{2m}) \geq 2m + 2$;
- all fifteen differences of the six systems of Section 9.4, the readings and witness of the pair 4–5, the false identity "distinct parts ≥ 2 versus distinct parts ≥ 1" (`:nonzero`), and one difference entered with integer entries, as in $L = [-t * T + 1 \quad -t * T;\ -t * Y \quad 1]$;
- the least annihilator: B1–B3 of Section 9.5 (`:rational` without pole, with A of order 1 where the system gives order 2), and C3–C5, whose readings differ by a polynomial ($A = 1$);
- automata in the notation of Section 9.6: Euler's identity, a product identity that needs a shifted form ($\Gamma \neq 1$), and a pair proved after aligning the readings by a factor Y;
- the sum side of Example 7.5: the two hypergeometric elements, the fit with a pole of order 1, and the difference of numbers 7 and 8, proved with a remainder for number 8 only;
- Glaisher's identity for $k = 3$ (parts not divisible by 3 versus parts used at most twice), with both sides given by automata: period 3 and a dilation witness with $\lambda = 3$; and a false identity (parts not divisible by 3 versus distinct parts), which ends with `:nonzero`.

`families_quantumwz.jl` runs all 79 differences of the four families of Section 9.5, and `automata_quantumwz.jl` all differences within each list of a file in the notation of Section 9.6; it also reports whether the proved differences connect every list. `stress_quantumwz.jl` draws random automata over $\{X, Y\}$ and runs `attempt` on differences whose answer is known: two unrelated automata (a proof would be wrong whenever $S \neq 0$ is certain from β_N), and an automaton against its transpose, against itself with a state split in two, and against the transpose of its canonical realization ($S = 0$ by construction; a refutation would be wrong).

Limitations. Channels must have ratios in K (roots of the coefficient equations are sought in $\mathbb{Q}(t)$), the degree of C in `qhyper` is capped at 64, periods are tried up to $p_{\max}$, offsets up to $|\mu| \leq 6$, shifted forms over four windows, and stationary channels must cancel in pairs through a dilation witness. The enumeration of candidates in `qhyper` is exponential in the number of irreducible factors of q_0 and q_k; above 50 000 candidates it is skipped. A failing search tries every combination of readings for every period, so `:unknown` can take long; the keyword `time_limit` of `attempt` bounds the channel search (it is checked between progressions, and inside `qhyper` and `section_recurrence`). A fit has pole order at most 6, and its elements come only from q-hypergeometric solutions of the adjoint with ratios in K. The least annihilator is guessed

within the bounds of step 4 (at most 32 unknown coefficients, entries reconstructed from at most 512 evaluation points, and rational coefficients of height about 2^{30}, lifted from one prime), so a reduction of high order or degree is missed. All these bounds affect only completeness: a skipped search never produces a verdict, and a guessed annihilator is used only after it is proved.

Status. The module was run with Julia 1.12.7, Nemo 0.56.1 and AbstractAlgebra 0.50.4. The 53 tests pass in about five minutes, most of it for Glaisher's identity and the sum side. The 79 differences of Section 9.5 take three and a half minutes in all, the 813 of the first set of Section 9.6 17 minutes, and the 3237 of the second set 138 minutes on one core, 94 of them for the 29 that are not proved. Three runs of `stress_quantumwz.jl` with 40 random cases each (automata with 2 to 4 states) gave no wrong verdict and no exception: of the 120 random differences, 113 were refuted and 7 proved (equal marginals), and all 310 identities were proved, by the zero test or by a rational certificate without pole, the slowest in 27 seconds. Oscar itself could not be installed where this version was prepared. There, its two-generator PBW algebras were replaced by a small stand-in with the same interface: normal-ordered terms through `coefficients` and `exponent_vectors`, the relation $g_2 g_1 = t\, g_1 g_2$, and scalars accepted only from K or $\mathbb{Z}$. An exact reference implementation in Python, on Flint, was used to cross-check the series, annihilator and channel stages.

11 The decision question

Four assertions should be kept apart.

1. The formal boundary always lies in B^{hol} with an effective description (Theorem 4.2).
2. Whether the marginal vanishes identically, $C \equiv 0$, is decidable (Proposition 4.5). This settles the comparison of realizations, but $S(\Delta) = 0$ is weaker than $C \equiv 0$.
3. Existence of rational certificates, with or without a pole at $T = 0$, is decidable (Theorem 5.4); those of the form $b = P \cdot C$ correspond to the rational solutions of (15) for the least annihilator of C, which the implementation guesses and then proves (Proposition 5.5). Without a pole they are complete proofs, but they are not necessary for $S(\Delta) = 0$: for identities between infinite products every certificate has a pole at $T = 0$ (Proposition 6.2).
4. Proposed convergence certificates (contraction, channels) are checkable by finite exact computation, and enumerating them is a semi-decision procedure for $S(\Delta) = 0$ *within* those classes. The test of Section 7.4 depends only on the sequence β, and with the readings of Section 9 it may use the partial sums of any reading of each block. It succeeds whenever, for some combination of readings, the marginal vanishes or some section of β is a sum of q-hypergeometric terms with ratios in K whose stationary parts cancel in pairs through dilation witnesses (within the search bounds); with the remainders of Section 7.5, it suffices that the partial sums of each block are such a sum up to a regular combination of themselves that tends to 0. Completeness beyond this class is not established; the theta series of Section 9.6 lies outside it.

A complete, effectively checkable certificate language for convergence would decide $S(\Delta) = 0$: enumerate certificates while computing coefficients of $S(\Delta)$ until either a certificate verifies or a nonzero coefficient appears. Equivalently, the problem is decidable iff there is a computable $N(\Delta)$ such that $t^{N(\Delta)} \mid S(\Delta)$ implies $S(\Delta) = 0$. Neither decidability nor undecidability is established here.

A Holonomic series do not form a field

Let $E(Y) = \prod_{j\geq 1}(1-t^jY) \in \widehat{B} \cap K[[Y]]$. Then $E(tY) = E(Y)/(1-tY)$, $E(0) = 1$, and $(1-t^n)[Y^n]E = -t^n[Y^{n-1}]E$, so the coefficients lie in K. Put $g = 1 + E \in B^{\mathrm{hol}} \cap \widehat{B}$; indeed $\big((T-t)((1-tY)T-1)\big)\cdot g = 0$. Since $g \equiv 2 \pmod{t}$, $g^{-1} \in \widehat{B}$.

E is transcendental over $K(Y)$. Otherwise $K(Y,E)$ is a finite extension of degree $r \geq 1$ stable under $\sigma = T$, since $\sigma(E) = E/(1-tY)$ and $\sigma^{-1}(E) = (1-Y)E$. The norm $a = N_{K(Y,E)/K(Y)}(E) \in K(Y)^\times$ then satisfies $a(tY)/a(Y) = (1-tY)^{-r}$. The left side has equal numerator and denominator degree, the right side does not: a contradiction.

g^{-1} is not holonomic. With $P_n = \prod_{j=1}^{n}(1-t^jY)$ we have $T^n \cdot g^{-1} = P_n/(P_n+E)$. As rational functions of the transcendental E over $K(Y)$ these have pairwise distinct simple poles $E = -P_n$, hence are linearly independent over $K(Y)$. Thus

$$g \in B^{\mathrm{hol}} \cap \widehat{B}, \qquad g^{-1} \in (F^{\mathrm{hol}} \cap \widehat{B}) \setminus B^{\mathrm{hol}}.$$

References

[1] S. Garoufalidis and T. T. Q. Lê, *A survey of q-holonomic functions*, Enseign. Math. 62 (2016), 501–525. arXiv:1601.07487.

[2] J. Middeke, *Denominator bounds and polynomial solutions for systems of q-recurrences over $K(t)$ for constant K*, ISSAC 2017. arXiv:1705.04188.

[3] NIST Digital Library of Mathematical Functions, *Integer partitions: other restrictions*, §26.10. dlmf.nist.gov/26.10.

[4] S. A. Abramov, *Rational solutions of linear difference and q-difference equations with polynomial coefficients*, Proc. ISSAC'95, ACM, 1995, 285–289.

[5] S. A. Abramov, P. Paule and M. Petkovšek, *q-Hypergeometric solutions of q-difference equations*, Discrete Math. 180 (1998), 3–22.

[6] M. Petkovšek, H. S. Wilf and D. Zeilberger, *A=B*, A K Peters, 1996.

[7] OSCAR documentation, *Creating PBW-Algebras*, `https://docs.oscar-system.org/stable/NoncommutativeAlgebra/PBWAlgebras/creation/`.

[8] OSCAR documentation, *Solving Linear Systems*, `https://docs.oscar-system.org/stable/AbstractAlgebra/how_to_linear_solve/`.